

GENERAL PSYCHOLOGY

PSYCHOLOGY: AN OVERVIEW

WHAT IS PSYCHOLOGY?

1. Definition - The book defines psychology as the study of behavior and cognitive processes. What does that mean? Does it mean the same thing to everyone? NO!!
2. Different Definitions of Psychology - Different people in the past and different people today hold a wide variety of views as to what psychology is and what it should study.
3. Wilhelm Wundt - Wundt founded the first psychological lab in 1879 at Leipzig in Germany. He saw psychology as the study of conscious experience. He did this through introspection, asking the person to describe what was doing on in his mind. His student Edward Titchener brought the idea of structuralism to the United States. It was way too dependent on verbal communication and truthfulness of participant. Introspection was an attempt to study psychology as a chemist would study substances.
4. William James - The first American psychologist. He was more interested in how the mind functions. The mind is a whole and you miss the point when you try to break it down into its parts. Develop functionalism as an answer to the problems encountered in introspection or structuralism.
5. John Watson - Watson said "You can't see the mind so how can you study it?" All we can see is observable behavior and that is what psychology should study. Skinner is also widely known for this view.
6. Mary Calkins - First woman to make a career in psychology. But because of prejudices against women in 1895, she never received a Ph.D. The first woman psychologists with a Ph.D. was Margaret Washburn.

MODERN APPROACHES TO PSYCHOLOGY

1. Overview - Today most psychologists believe overt behavior is central to the field. What leads to behavior is debated however. Therefore there are a variety of perspectives on what is the focus of study.
2. Cognitive Perspective - Cognition means thinking. The cognitive perspective holds that to understand how people behave you have to understand how they think, remember, and decide. That is what determines how they act.
3. Physiological Perspective - Our brain is a complex biochemical organ that affects how we think and how we behave. To understand behavior you must understand the roots of behavior, the working of the brain and nervous system.
4. Sociocultural Perspective - Behavior is strongly influenced by social and cultural factors. To fully understand behavior you have to take into consideration the social and cultural environment in which it occurs.
5. Psychodynamic Perspective - This is based on the work of Sigmund Freud. Freud saw psychology as the study of unconscious because he believed unconscious motivation controlled behavior.
6. Behavioural Perspective - Probably the most dominant perspective today. This is the continuation of Watson's idea that overt behavior is the most important thing to study. A combination of this and the cognitive perspective is also very popular.

TRAINING

1. Psychologist vs. Psychiatrist - Many people get these confused. They are similar in that both may work in the diagnosis and treatment of mental disorders. They are different in that a psychiatrist is a medical doctor who may prescribe medication. A psychologist generally has a Ph.D. and cannot prescribe medication.

2. Degrees in Psychology - The lowest degree is a Bachelor's degree. There is not much you can do with that. The second level is a Master's degree. This is the lowest level that allows you to practice but you generally need some type of supervision. Most practicing psychologists have a doctoral degree (Ph.D. or Psy.D.). This is the only level for full licensure and independent practice.

PROFESSIONS IN PSYCHOLOGY

1. Clinical Psychology - This is what most people think when they hear the word psychologist. They diagnose and treat mental disorders.

2. Counselling Psychology - The book says they treat people who do not have mental disorders but in reality they do the same thing as clinical psychologists although they are less likely to work with people with severe psychopathology.

3. School Psychology - This is the third of the three main applied areas. They diagnose and treat children with learning, behavioural, and emotional problems with an emphasis on the school environment.

4. Developmental Psychology - They study how people change physically, cognitively, and socially over the lifespan.

5. Educational Psychology - They study psychological principles within an educational environment. They are different from school psychologists because they generally do not work with individual children but focus more on instruction, techniques, etc.

6. Cognitive Psychology - Their focus is on thinking, reasoning, language, and decision making.

7. Industrial/Organizational Psychology - Studies aspects of behavior in work settings such as personnel selection, employee evaluation, work motivation, and leadership.

8. Psychobiology - Also called physiological psychology. They are interested in the brain and how it works.

9. Social Psychology - They are interested in social behavior. How do we act and react based on social information we receive.

10. Experimental Psychology - This covers almost all of the basic areas such as perception, learning, motivation, and animal behavior.

THREE MAJOR PHILOSOPHICAL ISSUES IN PSYCHOLOGY

Free-will versus Determinism. Is our behavior pre-determined for us or do we have a say in it?

The Mind-Body Problem. Is the mind (the psychological) separate synchronous or related to the body (physical)? The roots are in philosophy, but a number of early psychologists studied this problem.

Nature versus Nurture. Is behavior due to heredity (nature) or environmental influences (nurture)? A number of evidence goes both ways.

METHODOLOGY AND STATISTICS

METHODOLOGY

1. Definition - Methodology simply refers to the methods we use to conduct an investigation.
2. Systematic Research - Research in psychology is not haphazard. Following the basic principles of science there is some systematic way all research is conducted. Usually this means the psychologists approach a problem from a theoretical perspective, they arrive at some question they want to answer which is called a hypothesis, and then perform a study to answer the question.
3. Naturalistic Observations - This is the simplest way we study behavior. You simply observe the behavior in its natural environment. This is frequently informal is usually the first step to allow you to get a better understanding of the behavior which allows further, more in-depth investigation.
4. Case Study - This is the gathering of detailed information on a specific individual. It is the technique used by Freud. It is difficult; however, to generalize past the person you are studying.
5. Surveys - Surveys ask a large number of people questions and are most frequently used to get an idea of the population's attitudes toward something such as who they are going to vote for. They are only accurate, however, if the people surveyed are representative of the population as a whole.
6. Correlations - This investigates the degree of relatedness between two variables. The range is from -1 to +1. A positive correlation indicates that a high score on one variable is associated with a high score on the other variable. For example, height and weight are positively correlated. As you get taller you tend to weigh more. A negative correlation means that a high score on one variable is associated with a low score on the other. For example, amount of brain damage and short-term memory. The more brain damage you have the poorer your short-term memory. Correlation does not imply causality. For example, there is a high correlation between umbrellas being open and the number of puddles on the ground. That does not mean the umbrellas cause puddles.

EXPERIMENTATION

1. Causation - If you want to determine if one variable causes another to happen you have to conduct an experiment. An experiment systematically alters one variable to see its effect on another variable.
2. Independent Variable - The variable that we alter systematically is referred to as the independent variable.
3. Dependent Variable - This is the outcome variable, the one we measure to see how it is affected by the independent variable.
4. Example - You are interested in seeing the immediate effects of watching TV violence on aggression. You show one group a violent film (Terminator II) and another group a nonviolent film (The Sound of Music). You then put each in a play situation and see how much pushing the child engages in. The IV is the film and the DV is the amount of pushing.

How Some Experiments Go Wrong

There are a number of events that can occur before, during and after an experiment that can have a negative impact on the psychological research study. Poorly planned studies will yield results that are useless, misleading or worse. Here are some of them.

1. Demand Characteristics: the participants will not behave normally during the experiment. Instead he or she will try to guess what the experimenter is trying to do or study and either "help" or "hinder" the researcher. For example, if a study is titled: "Weeping and Crying." the participants may get an idea about what the study is about and act accordingly.

2. Experimenter Bias. The researcher can have his or her own perception of each participant. If a researcher expects "smart" subjects to do well, those subjects generally end up doing better than other's that were not expected to do well.

3. Dropout. The participants who decide to drop out of the study may be doing so because they possess certain characteristics that are not suited to the study.

To counter some of the problems involved, the primary investigator can design the study so that the experimenter and/or the participant are unaware of the purpose of the study. When just the participant or just the experimenter does not know the purpose, it is called a "single-blind" study. If both experimenter and participants do not know it is called "double-blind."

STATISTICS

1. Utility - We use statistics to determine the probabilities of differences in the DV are large enough so we are confident they are not due to chance. We can never be totally sure but we may be able to say we are 95% or 99% sure that differences are not due to chance.

2. Descriptive Statistics - These do not tell us anything about probability but only allow us to take a large amount of data and put it in some understandable form. These are grouped in two major forms: Measures of Central Tendency tell us about the center of our scores and Measures of Dispersion tell us how spreads out our scores are.

3. Measures of Central Tendency - Mean is simple the average score. This is calculated by adding up all the scores and dividing by the total number of scores. The mode is the most frequently occurring score. The median is the middle most score.

4. Example - Scores on an exam. Ten people in the class.

Scores	Statistics
83	Total = 810
79	
75	
98	
86	Mean = 81
91	
61	Mode = 75
80	
75	Median = 81
82	

5. Measures of Dispersion - Range is the difference between the highest and lowest score. In this case $98 - 61 = 37$. We usually use two other measures of dispersion:

Variance which is the average squared distance between each score and the mean and Standard Deviation which is the average distance between each score and the mean.

6. Normal Curve - Decisions on probability are based on the normal curve (Draw on board). We assume that most behaviors and traits are distributed normally within the population. IQ is an example. The further you deviate from the middle, the more confident you are the person is abnormal.

7. Inferential Statistics - These are the tests we use to determine if groups differ significantly, that is, not by chance. In other words, we can infer a difference based on probability. Don't need to know any details.

BIOLOGICAL BASES OF BEHAVIOR

OVERVIEW

1. Influence of Body on Behavior - No one will argue that the body and its mechanisms have a strong influence on behavior. This is especially true of the brain.

2. Phineas P. Gage - Back in the 1800's there was a man named Phineas Gage who was working in the construction of a railroad. His job was to use explosives to blow away ground so they could lay track. He was generally considered a kind and gentle man. One day an accident happened and the tamping rod was accidentally blow clean through his head. Surprisingly he didn't die but he did show a complete personality change. He became a drunk, was violent, and completely irresponsible. This led to an interest in how the brain controls behavior and personality.

3. Physiological Psychology - This field of study has traditionally been called physiological psychology but is now more frequently called psychobiology, biopsychology, or neuroscience. The focus of study is generally how the brain controls all aspects of our behavior.

NEURONS

1. What Are They? - Neurons are cells within our nervous system. They are specialized for sending and/or receiving information.

2. Structure - Neurons consist of three basic parts. The cell body is the main part or center of the neuron. The axon is a tail like structure that carry information away from the cell body. The dendrites (one or more) are usually small appendages that carry information toward the cell body.

3. Myelin Sheath - Many axons, especially the longer ones are covered by a fatty substance called myelin. It is interrupted by small gaps called Nodes of Ranvier which allow for impulses to be sent more quickly. At the end of the axon is an axon terminal that comes close to another cell in an area known as the synapse.

4. Action Potential - Neurons transmit messages in the form of electrochemical impulses called an action potential.

5. Neurotransmitters - When the action potential reaches the axon terminal a chemical known as a neurotransmitter is released into the synapse. They travel across the synapse until they reach a receptor (usually in the dendrites of another neuron) and they cause that cell to respond according to the message received. Students don't need to know any specific neurotransmitters.

THE NERVOUS SYSTEM

1. Definition - The nervous system connects our entire body and allows communication that controls our behaviors and bodily functions. There are two major systems: the Central Nervous System and the Peripheral Nervous System.
2. Central Nervous System - This consists of the brain and spinal cord. This is the ultimate control center for all human behavior.
3. Peripheral Nervous System - This consists primarily of nerves that conveys signals from the body's sensory receptors to the CNS and transmits messages back from the CNS to muscles or glands.
4. Divisions of Peripheral Nervous System - The PNS is divided into two parts. The Somatic Nervous System connects the CNS with voluntary muscles throughout the body, those muscles we can control. The Autonomic Nervous System controls visceral muscles and glands. Those muscles we don't have voluntary control over such as the heart.
5. Divisions of Autonomic Nervous System - The ANS is divided into two parts. The Sympathetic Nervous System promotes energy expenditure, especially in emergency situations. It stimulates heart rate, BP, flow of blood to muscles, etc. to get you ready for action. The Parasympathetic Nervous System dominates under conditions of relaxation and tends to conserve energy.

STRUCTURE OF THE BRAIN

1. Very Complex - The human brain is the most complex structure ever discovered or created by humans. There are approximately 10 billion nerve cells in the brain and each one has potential contact with every other one. There is, however, no relationship between brain size and intelligence. What relate to intelligence are the convolutions (fissures, wrinkles, or folds) in the brain. No animal has as many convolutions as humans except the porpoise.
2. Oxygen Consumption - The brain needs oxygen to survive. The brain contributes about 1/50th of your body weight yet consumes 25% of your oxygen. Deprivation of oxygen to the brain will lead to you passing out in 15 seconds and brain damage within 4 minutes.
3. No Pain Receptors in Brain - There are no pain nerves in the brain. If you have a headache it is usually due to dilation off the blood vessels that surround the brain.
4. Brain Stem - Our basic body processes are mediated in the brain stem, the portion of the brain just above the spinal cord. The Medulla regulates vital body functions such as breathing and heartbeat. This is why people that are called brain dead can still survive. The Pons deals with sensory and motor information and regulates sleep, arousal, etc.
5. Cerebellum - This is just behind the brain stem and it coordinates voluntary movements, regulates posture and equilibrium.
6. Midbrain - Located near the end of the brain stem. This contains primitive centers for vision and hearing and plays a role in the regulation of visual reflexes.
7. Hypothalamus - This is a small structure that has a tremendous effect on our behavior. It regulates the autonomic nervous system and influences behaviors such as eating and aggression as well as our sexual organs.

8. Cerebral Cortex - This is what sets humans apart from other animals by its size and this is the center for all higher mental processes. It consists of two hemispheres and four distinct areas which are called lobes.

(a) Frontal Lobe - This is the area closest to the face. The function of most of it is unknown but it is believed to have some effect on behavior. This is the area that Phineas P. Gage had damaged. This led to psychosurgery, the removal of this part of the brain. It also contains the motor cortex, the area concerned with control of body movements

(b) Parietal Lobe - This is located just behind the frontal lobe across the central fissure. This contains the somatosensory cortex and is the primary area for reception and interpretation of information from the skin senses (touch, temperature, etc.).

(c) Occipital Lobe - This is located in the back of the head and it is the primary visual area in the brain. It receives input from the eyes.

(d) Temporal Lobe - Located along the side of each hemisphere and it is primarily concerned with hearing and receiving input from the ears.

THE ENDOCRINE SYSTEM

1. Endocrine System - Our body and behavior are controlled not only through neural impulses from the brain but also through a slower chemical system called the endocrine system.

2. Hypothalamus - The Hypothalamus has control over endocrine glands located throughout the body. The messages are sent to the gland via chemical substances called hormones. These hormones affect sexual arousal, emotional responses, motivation, physical growth, and metabolism.

3. Pituitary Gland - This is called the master gland in the body and it is what receives information from the Hypothalamus and then releases hormones to affect other parts of the body.

PRINCIPLES OF HEREDITY

1. Conception - Conception occurs when a sperm cell from the father penetrates and unites (fertilizes) with the ovum (egg cell) from the mother. The egg is about 90,000 times as heavy as the sperm. Women have finite number of eggs (300-500), men infinite number of sperm. For example, a teaspoon of sperm could father everyone alive on earth today.

2. Chromosomes - In the nucleus of each cell there are threadlike entities called chromosomes and on these there are genes which contain the genetic code. Each cell of normal people contain 46 chromosomes (2 sets of 23). The sperm and the egg each have 23 chromosomes so that when they unite the fertilized egg gets one set of chromosomes from the mother and one from the father.

3. Genotype & Phenotype - Genotype is simply the genes that you have. Phenotype is how those genes are interpreted into behavior or characteristics.

6. Sex Chromosomes - Of the 23 pairs of chromosomes, 22 are autosomes, that is they are possessed equally by males and females. The 23rd pair is the sex chromosome and differs in males and females. The female has 2 X chromosomes (XX) and males have an X and a Y (XY). As the mother can only pass on an X chromosome the sex of the child is determined by the father.

7. Sex-Linked Characteristics - The fact that some genes on the X have no counterpart on the Y results in sex-linked characteristics in males. These are result of genes on X that only show self in presence of Y. Examples include hemophilia, color blindness, baldness, etc. Also, there is a higher rate of miscarriage. infant mortality, and childhood deaths in males partly because of their vulnerability to sex-linked disorders.

CHROMOSOMES ABNORMALITIES

1. Down Syndrome - Called trisomy 21 because child has an extra 21st chromosome. Characterized by physical and mental retardation and a distinctive physical appearance. Also have greater susceptibility to leukemia, heart disorders, and respiratory infections which often lead to early death. Occurs more frequently in older mothers (over 35). This is the most common of the chromosomal abnormalities.
2. Sex Chromosome Abnormalities - Many chromosomal abnormalities involve the sex chromosomes, usually you have too many.
3. Turner's Syndrome - (XO) Born with a single X chromosome. Usually have normal intelligence but are small in stature, webbed neck, stubby fingers and toes, broad chest and small breasts. Personality is usually pleasant, relaxed, and docile. Sterile at puberty and they don't develop secondary sex characteristics. Can be treated with hormones which develops secondary characteristics but they remain sterile.
4. Triple-X Syndrome - Also called "super female" syndrome. Have 3 X chromosomes. They are normal in appearance and fertile. Slightly below average in intelligence.
5. Klinefelter's Syndrome - These are males born with one or more extra X chromosome (XXY, XXXY). Appear normal until puberty when they develop female secondary sexual characteristics (breasts, etc.). They are sterile and about 20-30% are mentally deficient with retardation being more severe with the more X chromosomes.
6. Supermale Syndrome - (XYY) Males with extra Y chromosome. Tend to be taller, have acne, and slightly below average IQ. There has been some suggestion that they are more likely to be aggressive and criminals. This is not warranted. Although there may be higher rate of them in prison than in normal population, their rate is similar to XY low IQ males.

GENETIC ABNORMALITIES

1. Cause - Many serious disorders are caused by a single defective gene. Could be result of mutations in genes which can be caused by high temperature, radiation, drugs, or sometimes happen spontaneously. Also caused by recessive genes.
2. PKU (phenylketonuria) - Child is unable to metabolize phenylalanine. If left untreated it results in MR, hyperactivity, and a peculiar odor. Can be treated via diet. Caused by a recessive trait.
3. Tay-Sachs - A degenerative disease of the nervous system that kills its victims, usually by age 3. A recessive gene so both parents must be carriers. Affects Jewish children of Eastern European ancestry.
4. Cystic Fibrosis - Also recessive trait. Lack enzyme to prevent mucus from obstructing lungs. Rarely survive beyond adolescence.

5. Huntington's Chorea - Dominant gene. Causes a gradual deterioration in nervous system, leading to death. Usually doesn't appear until after age 40. It can therefore be passed on from generation to generation.
6. Genetic Counselling - When parents have a history of genetic disease in family or already had a defective child may want find out odds of child being defective. Counselling may just tell them the odds, can have their genes screened, or may be procedure on mother if pregnant to determine if fetus OK. Amniocentesis, procedure where needle inserted in abdomen, amniotic fluid is withdrawn, and chromosomes of baby examined.

SENSATION & PERCEPTION

SENSATION & PERCEPTION

1. Sensation - Sensation refers simply to your contact with the external world through your sensory receptors. These are the inputs received via our sensory receptors.
2. Perception - This refers to how we interpret and organize the information we receive from the senses. It is our conscious experience of what our senses are telling us.

SENSATION

1. Stimulus - A stimulus is any form of energy to which our senses respond. For example, the light that hits our eyes would be considered a stimulus.
2. Sensory Receptors - In certain areas of our body we have specialized receptor cells that are programmed to respond to specific stimuli. They convert (or transduce) the physical properties of a stimuli into neural impulses that our brain can understand.
3. Sensory Deprivation - Our bodies are prepared to receive a great deal of sensory input. When we are deprived of sensory input our brains sometimes play a trick on us and tell us we are receiving information when we are not. This is usually in the form of hallucinations.
4. Absolute Threshold - Not all stimulation is received and translated. For example, a dog can hear sounds you cannot. The absolute threshold is the smallest amount of a stimulus that we can detect 50% of the time.
5. Signal Detection Theory - Some scientists dispute the existence of absolute thresholds. They propose a theory called signal detection theory. This holds that there are no absolute thresholds but rather, detection of a stimuli depends on the energy of the stimulus and the costs and benefits to you of detecting that stimulus. For example, if you are in a crowded room you do not hear what everyone is saying even though all sounds may meet a minimal threshold for hearing. You hear only those sound you deem important enough.
6. Difference Threshold - This is the amount two stimuli must differ for you to notice that they are different. This is illustrated by listening to your stereo. How much do you have to turn it up for you to notice that it is louder. Basically, the lower the intensity the easier it is to notice a difference. This is referred to a Weber's Law which states that the smallest detectable difference is a function of the intensity.
7. Subliminal Perception - There has been some interest in our ability to respond to stimuli that is below our threshold of conscious perception. This is very controversial and there is no agreement that it occurs. Some people believe however that there are satanic messages in rock albums.

8. Sensory Adaptation - Following prolonged stimulation, all sensory systems display adaptation which refers to an adjustment that decreases sensitivity. An example would be washing dishes. When you first put your hands under hot water the water seems very hot but gradually it does not seem that hot anymore.

VISION

1. Structure of the Eye - The human eye has five main parts: cornea, pupil, iris, lens, and retina. The cornea is the transparent outer layer through which light passes. The pupil is the opening just behind the cornea through which light enters the eye. The iris is the colored part of the eye that adjusts the amount of light that enters the eye by constricting or dilating the pupil. The lens is just behind the pupil which bends the light to focus it on the retina which is at the back of the eye and converts the light into neural impulses that our brain can interpret.

2. Receptors on Retina - There are two types of receptors cells on the retina. Rods are responsible for responsible for light and dark reception and are more sensitive to movement. There are about 100 million rods on the retina. Cones are centered on the fovea and are responsible for color vision and feature detection.

3. Color Vision - There are two main theories that are used to explain color vision. Trichromatic Theory suggests that there are three types of cones each sensitive to different wavelengths. Some respond to red light, others green, and others blue. When each is stimulated you see that color, when more than one is stimulated you see the color between them. This has weaknesses. For example, it doesn't explain negative afterimages. This is when you stare at one color for a long time, when you look away you see the complementary color. Opponent-Process Theory states that there are three antagonistically organized systems. Two systems are composed of opposite colors (red-green & yellow-blue). When one is stimulated, the other is inhibited. When both are stimulated they cancel each other out. The other system responds to black and white. Experimental evidence suggests that color vision involves both systems.

HEARING

1. Outer Ear - This is the external, projecting portion of the ear. It consists of the pinna, the visible part and the auditory canal.

2. Middle Ear - This is separated from the outer ear by the eardrum and contains three small bones (hammer, anvil, & stirrup).

3. Inner Ear - Also called the cochlea which is a fluid filled chamber attached to the middle ear by a thin membrane called the oval window.

4. Perceiving Sound - Sound waves enter the ear, goes down the auditory canal and strikes the eardrum. The eardrum moves which causes vibration to the bones in the inner ear. They amplify the sound waves and apply pressure to the oval window which causes vibration of the fluid in the cochlea. There are hair cells in the cochlea which move with the fluid and convert the sound to neural signals which go to the brain.

5. Types of Sounds - Not all stimulation to the ear produce the same perception of sound. We perceive pitch, which is the highness or lowness of the sound. This is determined by the frequency of the sound wave. We also perceive the loudness of sound which is a function of the magnitude of the sound waves. Finally we perceive

timbre which is the quality of the sound and is a function of the complexity of the sound wave.

6. Pitch Perception - There are two main theories on how we perceive different pitches. Place Theory suggests that sounds of different frequencies stimulate different areas of the cochlea. Frequency Theory suggests that sounds of different frequencies cause different rates of neural firing. In actuality, both theories explain pitch perception. Frequency theory explains low frequency sounds and place theory explains high frequency sounds.

SKIN SENSES

1. Touch - There are sensory receptors in the skin that allow us to perceive touch. Sensitivity varies depending on the part of your body. Certain areas of the body have more receptors and a larger part of the brain devoted to them. Fingers and mouth are the most sensitive.

2. Warmth and Cold - Physiological zero is your skin temperature. If you touch something of this temperature you feel no temperature. There are warmth and cold receptors. Hot is perceived by simultaneous activation of warmth and cold receptors.

3. Pain - Pain has no specific receptors. Pain is caused by damage to bodily tissues which release chemical substances that stimulate pain neurons.

CHEMICAL SENSES

1. Smell and Taste - These are closely related senses. The principle function of smell is to warn you about potentially toxic substances.

2. Smell - To smell, molecules enter the nose and are dissolved in nasal tissue. This brings them into contact with receptor cells which send the information to brain. Humans have about 10 million receptor cells. Dogs, however, have about 200 million.

3. Taste - There are four basic categories of taste: Sweet, sour, salty, and bitter. There are receptors for these on different parts of the tongue. The sensory receptors for taste are located in the taste buds.

BALANCE, POSTURE, AND MOVEMENT

1. Balance - This is called the vestibular sense. It is controlled by three semi-circular canals within the cochlea in the inner ear.

2. Kinesthesia - This is your sense of body movement and position. It works with your vestibular and visual senses to maintain balance and equilibrium. The receptor cells for this sense are found in the nerve endings in and near joints, muscles, and tendons.

PERCEPTION

1. Definition - Perception is an organism's awareness of objects and events in the environment brought about by stimulation of the sense organs.

2. Attention - We cannot be aware of all stimulation we receive at any time. You would be inundated with too much information. We selectively attend to certain aspects of our environment while relegating others to the background.

3. Perceptual Organization - This is the process by which the brain structures the input from our sensory receptors.

4. Gestalt Psychology - The Gestalt view of perception is that all humans organize or group information into meaningful patterns called gestalts. In other words, we view the whole rather than the parts. We tend to organize data by its proximity (how close together), continuity (is it a continuous grouping?), and similarity (the likeness of stimuli). We also view things in terms of figure and ground. This is our tendency to view something as the region that represents the object and its background.

5. Perceptual Constancy - This is our tendency to perceive objects as having certain constant, unchanging properties. Size Constancy is our tendency to perceive something as a certain size even though it is further away and makes a smaller image on the retina. Shape Constancy refers to our tendency to perceive a shape as the same even though the image on the retina may differ. The book gives the example of a coin. We see it as round even though we rarely see it this way.

6. Illusions - Sometimes our perceptions yield false interpretations of physical reality. Use transparencies to illustrate illusions. Moon Illusion: Why does the moon look bigger on horizon than when it is high in the sky? The image on the retina is the same. We know it is far away and when it is on the horizon we have visual cues such as trees and houses that trick us. We are still not quite sure if this is it however.

7. Depth Perception - This is usually derived from binocular disparity, the brain using the slight differences in information received from each eye to perceive depth. There are also cues that do not require the cooperation of both eyes. These are called monocular cues. Motion Parallax is the differences in the relative movement of retinal images that occur when we move or change position. Objects closer appear to move faster. Interposition is when one object partially blocks the view of another object creating the perception that the second object is farther away. Linear Perspective is the apparent convergence of two parallel lines in the distance. Relative Size is the relationship between the size of an image projected on the retina and the distance of the object from the observer. The larger the retinal image, the closer the object appears to be. Texture Gradient is the closer an object is to you, the more texture you can see. As an object gets farther away it appears smoother.

LEARNING THEORY

CLASSICAL CONDITIONING

1. Discovered the principle of classical conditioning by accident.
2. Pavlov wanted to understand how a dog's stomach prepares to digest food when something is placed in the dog's mouth.
3. He noticed that the mere sight or smell of food was enough to get the dog salivating - investigated how this worked.

ELEMENTS OF CLASSICAL CONDITIONING

1. UCS ----->UCR

(food) (salivation)

CR-----|CS

(tone) - neutral

2. Pavlov chose the tuning fork as a neutral stimulus - it initially had nothing to do with the response (salivation).

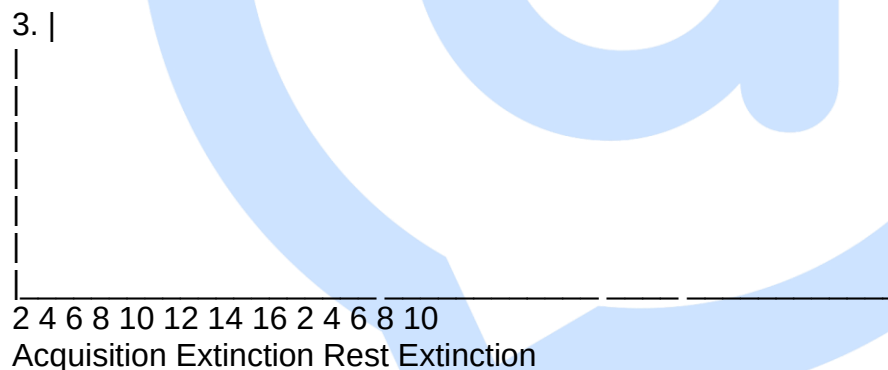
3. By pairing the neutral stimulus with the food he found that the tone alone could elicit salivation.
4. Generally thought of as involving responses the individual has little control over such as autonomic nervous system responses - salivation, sucking in infants, heart rate, and blood pressure.

CONCEPTS OF DESCRIBING CLASSICAL CONDITIONING

1. Unconditioned Stimulus (UCS) - the environmental factor (stimulus) that naturally brings about a particular behavior (response).
2. Unconditioned Response (UCR) - the unlearned, automatically occurring reaction brought about by an unconditioned stimulus.
3. Conditioned Stimulus (CS) - is a neutral stimulus that becomes capable of eliciting a particular response through being paired during training with a UCS.
4. Conditioned Response (CR) - is a response aroused by some stimulus other than the one that automatically produces it.

OTHER PRINCIPLES

1. Extinction - Pavlov discovered that if he stopped presenting food after sounding the tuning fork, the sound gradually lost its effect on the dog - it no longer elicited salivation. The association between the two lessened.
2. Spontaneous Recovery - He also found that if he gave the dog a rest from the apparatus after extinction and later struck the tuning fork again the dog salivated again. If pairing not made with UCS however, second extinction is quicker. Also successive spontaneous recoveries not as strong.



4. Stimulus Generalization - Not only the conditioned stimulus under which a response was first learned can elicit the response (CR) but also a range of similar stimuli. *Example:* Pavlov - not only tuning fork but also maybe hitting the edge of a glass.
5. Stimulus Discrimination - The ability to respond differently to two or more stimuli that are similar yet distinct.
6. Higher-Order Conditioning - After a secondary conditioned stimulus is paired with the primary stimulus a number of times, it may take on the functions of a primary stimulus.

Example: Pavlov's experiments have tone eliciting salivation, now let's pair tone with a light- after a while the light alone will elicit salivation.

EXAMPLES

1. Fear - Give example.

Watson & Raynor (1920) - demonstration of classical conditioning to emotional responses in humans. 9 month old child - Little Albert - beginning of experiment white rat did not elicit fear, loud noise did elicit crying and distress. Paired the loud noise with the rat and after a while the rat elicited fear in Little Albert.

What is UCS? (loud noise)

What is UCR? (fear response to loud noise)

What is CS? (rat)

What is CR? (fear of rat)

What would it be if Albert also became afraid of cats? (stimulus generalization)

What if Albert was not afraid of gerbils? (stimulus discrimination)

This is one way phobias can develop.

Most Common Phobias

acrophobia - fear of high places

agoraphobia - fear of open places

claustrophobia - fear of closed places

gynephobia - fear of women

hydrophobia - fear of water

mysophobia - fear of dirt

ophidiophobia - fear of non-poisonous snakes

Other Fairly Common Phobias

ailurphobia - fear of cats

algophobia - fear of pain

astraphobia - fear of lightning

hematophobia - fear of blood

monophobia - fear of being alone

nyctophobia - fear of darkness

ocholophobia - fear of crowds

pathophobia - fear of disease

pyrophobia - fear of fire

syphilophobia - fear of syphilis

xenophobia - fear of strangers

zoophobia - fear of animals or a specific animal

More Exotic Phobias

emetophobia - fear of vomiting

enosiohobia - fear of committing an unpardonable sin

entomophobia - fear of insects

eosophobia - fear of dawn

ergasiophobia - fear of work
erthyrophobia - fear of blushing in public
gamophobia - fear of marriage
gephyrophobia - fear of crossing a bridge
gymnophobia - fear of naked bodies
haphophobia - fear of being touched
heliophobia - fear of sunlight
homilophobia - fear of sermons
melissophobia - fear of bees
parthenophobia - fear of virgins
pnigophobia - fear of choking
taphophobia - fear of being buried alive

2. Taste Aversion - Let's say you go to a restaurant and you try escargot for the first time. You then get sick - will you eat snails again? (probably not). What if ate steak, baked potato, and snails? (aversion probably just to snails). Why? (snails are the novel stimulus).

OPERANT CONDITIONING AND REINFORCEMENT

Background

1. Definition - Type of learning in which the consequences of a behavior influence whether the organism will act in the same way in the future - the animal learns the relationship between his own behavior and a reinforcing or punishing stimulus.
2. Reinforcement - An environmental stimulus which is contingent on a response and increases the probability of a response.
3. Punishment - An environmental stimulus which is contingent on a response and decreases the probability of a response.

The ABCs of Behavior (Functional Analysis)

1. A = Antecedent - Stimuli happening before the behavior such as instructions or gestures.
2. B = Behavior - This is simply the act itself. The individual's response.
3. C = Consequence - This is the event that follows the behavior.
4. Example - A = A teacher asks a student to answer a question. B = The student answers the question. C = The teacher tells the student that he did good (reinforcement).

Four ways to Administer Contingent Consequences

1. Positive Stimulus presented (positive reinforcement) - presentation of a stimulus increases the probability of a response. Example? (give candy for sitting in seat)
2. Positive Stimulus removed (response-cost) - Should produce a suppression of the behavior. Example? (take away TV for talking back to mother)
3. Negative Stimulus presented (punishment) - Decreases the probability of the behavior. Example? (spanking)

4. Negative Stimulus removed (negative reinforcement) - Results in increase in probability of a behavior. Example? (kid wants a cookie, he cries until mother gives him one, mother's cookie giving behavior removes negative stimulus, crying, more likely to give cookie again to escape from crying - child however was positively reinforced for crying)

Reinforcement

1. Primary Reinforcement - Unconditioned. A stimulus that does not require an organism to learn its reinforcing qualities. Examples? (food, water, sex).
2. Secondary Reinforcement - Conditioned. A neutral stimulus that, through constant association with primary reinforcers, acquires its own reinforcing qualities. Examples? (money, grades)
3. Extinction - When we withhold a reinforcer than you should have extinction of the behavior it was reinforcing. Hope not always the case - when have added behavior through reinforcement, hope other things will have taken over reinforcing properties or it has become intrinsically reinforcing.
3. Bribery vs. Reinforcement - some say reinforcing someone for acting the way they should is Bribery but offering incentives is not necessary Bribery. Is a salary for working Bribery?

Factors Influencing Effectiveness of Reinforcement

1. Potency of the Reinforcer - Want it to be enough to be reinforcing but not too much or subject will satiate quickly and item will lose its reinforcing qualities. Example? (using edible reinforcer, M&Ms, is one enough but if give large bag may get sick of them).
2. Immediacy of Reinforcement - The greater the delay in administering the reinforcer the less the effect.
3. Verbalization - It helps verbalizing the connection between the behavior and the reinforcement to strengthen the connection.
4. Shaping - If behavior not in repertoire you will never have the opportunity to reinforce it. To get it in repertoire you reinforce successive approximations of behavior. Example: Nonverbal child getting them to ask for things instead of grabbing - first, grunting and pointing, then attempting to say cookie, then saying cookie, then cookie please, then may I have a cookie?

Schedules of Reinforcement

1. Continuous vs. Partial Schedules

Continuous - A reward is given every single time the response is elicited. Generally used only to establish a behavior. Easy to extinguish.

Partial - The subject is only occasionally rewarded for the proper response. More resistant to extinction.

2. Partial Schedules Either Ratio or Interval

Ratio - Based on the number of correct responses the organism makes between reinforcements.

Interval - Based on the amount of time that has elapsed between reinforcements.

3. Fixed or Variable Schedules

Fixed - regular schedule

Variable - irregular schedule

4. Four Possible Partial Schedules

A) Fixed-Ratio - Reinforcement depends on a certain amount of behavior being emitted, for example every fifth response is reinforced. Example? (a typist who gets paid after a certain number of pages are typed). Pattern - people generally work hard on FR schedules, pausing briefly after each reward. Problem - if the amount of work responses required before the next reward is large you will likely see low morale and few responses at the beginning of each cycle.

B) Variable-Ratio - The number of required responses varies around some average. Example? (Salesman who sells something to the second customer, then to the sixth, the eighth, etc. also gambler at a slot machine, the faster he puts money in machine the faster he will hit jackpot). Pattern: People tend to work at a high steady rate. No delay because reinforcement may come on next response.

C) Fixed-Interval - Reinforcement is given at a predetermined time no matter how many responses have been emitted. Example? (not many naturally occurring examples but studying just before a test may be one). Pattern: scallop effect - begin interval responding slowly but increase as interval progresses in anticipation of reward.

D) Variable-Interval - The time a reinforcer will be available varies around some average time. Example? (dialling a number when the line is busy, will get through but don't know when). Pattern: steady, moderately paced responses.

E. Which is Most Resistant to Extinction?

Variable interval and variable ratio most resistant to extinction with variable ratio being the best. Why? (never know when the next reinforcement is going to come). If you gradually increase the ratio or interval you can make the behavior even more resistant to extinction.

MEMORY

MEMORY

1. Definition - Memory is the capacity to retain and retrieve information.

2. Three Basic Memory Processes - There are three basic tasks of memory. Encoding is the acquisition component. This is transforming raw information into a form in which it can be entered into memory. Storage is the retaining of information over time. Retrieval is the ability to get encoded material back into awareness.

3. Information-Processing Approach - This is an approach to understanding human memory that emphasizes the encoding, storage, and retrieval of information. It uses a computer analogy.

TYPES OF MEMORY STORAGE

1. Sensory Memory - This is the momentary persistence of sensory information after stimulation has ceased. It begins to decay almost immediately if you don't attend to it. You are constantly being bombarded by sensory information so you must control the selection of what you encode. This is called selective attention.

2. Short-Term Memory - This is the memory system that holds limited amounts of information for relatively short periods of time. Information must be transferred from sensory memory to STM in order for it to be processed and become meaningful. This is

what we are consciously aware of at any time. The capacity of short-term memory is thought of as being 7 +/- 2 items. Information can be kept in STM indefinitely by rehearsing it. Short-term memory is thought to exist in a bio-electrical state and to be converted into long-term memory it must be incorporated into a chemical state.

3. Long-Term Memory - This is the memory system for large amounts of information over long periods of time. Its capacity is limitless yet retrieval of information is not always easy. For example, sometimes we know we know something but we can't remember it. This is referred to as Tip-of-the-Tongue Phenomenon. Long-term memory can be Episodic Memory which is an autobiographical record of what we see, hear, or do or Semantic Memory which is organized knowledge about words or symbols and the rules for manipulating them.

FORGETTING

1. Trace Decay Hypothesis - This is the simplest view of forgetting. It is simply that information entered into long-term memory fades or decays with the passage of time. Can you remember who was sitting next to you in second grade?

2. Interference - If it is not time that makes you forget things it may be interference of other things that are stored in memory. It can take two forms. Retroactive Interference is when information you are currently learning interferes with what is already in memory. Proactive Interference is when information previously entered into memory interferes with what you are trying to learn now.

3. Infantile Amnesia - You generally cannot remember much of anything that happened to you during the first 3 years of life. This could be because you lacked the brain structures for long-term memory at this age or lacked language which may be necessary to encode information into long-term memory. What memories people think they have from this period are usually things they were told by someone at a later time although you may be certain that you remember it. This is very controversial now because some people, after therapy, insist they remember infantile abuse. Most experts in the field discount this as being suggestions given to them by their therapist.

MEMORY RETRIEVAL

1. Types of Retrieval - Retrieval has two forms. Recognition is realizing that the information has been seen or heard before. A song for example. You may not know all the words but you know you have heard it before. Recall entails an active search of long-term memory to actually retrieve a specific piece of information.

2. Relearning - Relearning seemingly forgotten information can take less time than original learning because even though you have forgotten the details, there may be a trace left which facilitates storage and retrieval.

3. State Dependent Learning - Information which is learned in one physiological state may be difficult to retrieve when you are in another physiological state. If you meet someone when you are drunk you may not remember him sober but next time you are drunk you do.

4. Memory Distortion - Did you ever remember something one way and a friend remembered it totally different? We all possess schemas, or cognitive frameworks with which we view the world. Once a schema is formed it can influence the way you

encode, store, or retrieve information. You are more likely to remember things that support your beliefs.

5. Construction - Our memory is also affected by construction which is our tendency to fill in details when recalling past events or even remembering experiences you never had. You can't possibly remember all of the details so when you recall the experience you may put things in that you assumed happened or you wish happened.

6. Amnesia - Amnesia is the loss of memory usually from illness, accident, or drug abuse. Retrograde amnesia is inability to remember things that happened prior to the amnesia inducing event. Anterograde amnesia is the inability to store in long-term memory information that occurs after the amnesia inducing event.

COGNITION

THINKING AND REASONING

1. Cognition- Refers to all mental activities associated with thinking, knowledge, and memory.

2. Concepts - Are mental categories for objects, events, experiences, or ideas that are similar to one another in some respect. They help us understand the world by helping us represent it mentally. They may be represented in the mind in several ways such as through visual images.

3. Propositions - Thinking frequently involves relating one concept to another. Propositions are sentences that relate one concept to another and can stand as a separate assertion. For example, you could say "All professors are smart." You have a concept of what professors are and what smart is and you are linking them together.

4. Reasoning - Reasoning is simply drawing conclusions from available information. Formal reasoning derives conclusions from specific premises. This is what you may have had in a logic class. All men are pigs. John is a man. Therefore John is a pig. Everyday reasoning is less formal. It does not work on clear premises. You may say "John is a pig" but there is no clear premise that leads to that conclusion.

5. Errors in Reasoning - There are a number of things that can cloud our reasoning such as our emotions and beliefs. The confirmation bias is our tendency to focus heavily on evidence that confirms our initial preconceptions. For example, if you are in favor of abortion rights you may read only those reports supporting your belief. If you are anti-abortion you may read only those things that support that view. When you are exposed to the opposing viewpoint you may discount it.

6. Reasoning Effectively - The book gives some suggestions to reason more effectively. They are (1) Examine and test all premises. In other words, don't accept a premise until you have tested it, it may be flawed. (2) Guard against the confirmation bias. Recognize that you don't know everything and that there are other points of view. Actively seek out opposing points of view. (3) Recognize the role of emotions. Try not to make decisions when you are highly emotional.

LANGUAGE

Background

1. Definition of Language - A small number of individually meaningless signals (sounds, letters, and gestures) that can be combined according to agreed-on rules to produce an infinite number of messages.

2. Amazing - That's amazing. You start out with just a few meaningless sounds. From there you can make thousands of words which can be combined using rules (grammar) to produce an infinite number of messages.

3. Inventive - Language is also an inventive tool. What we say or hear in a given situation is not merely a repetition of what we heard in the past, we create novel utterances on the spot or a number of different topics. Yet, although you come up with these new things to say, other people can understand you as long as you adhere to the rules and conventions of the language you are using.

4. Learning Language - Language is very complicated yet children in all cultures come to understand and use language at a very early age. By 5, children not only understand most grammatical rules but can construct complex, adult-like sentences even though they have had no formal training in language. It appears effortless.

Theories of Language Development

A. Learning Theories

1. Reinforcement Model - In 1957 Skinner published a book entitled "Verbal Behavior" in which he argued that children learn to speak appropriately because they are reinforced for grammatical speech. Specifically, adults shape a child's language by selectively reinforcing those aspects of babbling that are most like adult speech, thereby increasing the likelihood that those sounds will be repeated. Once they have shaped sounds into words, they shape words into sentences, and then longer sentences, and so on until the child talks like an adult.

2. Brown, Cazden, & Bellugi (1969) - Study to test reinforcement theory. If reinforcement theory plausible, then parents should reinforce grammatical speech while discouraging ungrammatical utterances. They recorded conversations between mothers and their young children and found that in most cases found that mother's approval or disapproval depended on the truth value of the child's statement rather than on correct grammar. For example, when one child said "He a girl." (truthful but grammatically incorrect) the mother said "That's right." When the child said "There's the animal farmhouse" when pointing to a lighthouse (grammatically correct but untruthful) the mother corrected her. It appears parents pay little attention to their child's early grammar.

3. Imitation - This would be consistent with a behavioural perspective. They hear someone else say something and receive reinforcement so they are more likely to repeat it. It has been shown that children do learn the names for things by hearing others use those names. Problem comes with grammar. Bloom et al report that children do not readily imitate a grammatical rule until they have already used that principle at least once in spontaneous speech. Imitation may therefore help the child apply rules but not acquire them.

6. Conclusion - There appears to be a learning component in language development but it cannot fully explain a lot that goes on, especially in the area of grammar.

Nativist Perspective (Maturational)

1. Noam Chomsky - Took issue with Skinner and said operant conditioning could not fully explain language acquisition. There was no way children could learn proper

grammar by listening to adults who frequently don't talk very well. He thought there was an innate capacity; there is a biological program to acquire language.

2. Language Acquisition Device (LAD) - Sort of like a "black-box" for understanding the regularities of speech and fundamental relationship of words. It is just their unique, inborn capacity for language learning. Basically when they hear someone else use a new linguistic principle, they can detect it and apply it to their own speech.

3. Support for Nativist Position

(1) Brain Specialization - Major language centers are located in the left cerebral hemisphere. Studies show that from birth speech sounds elicit more electrical activity from left side of brain. Infants are also able to discriminate important contrasts (Ba - Pa) very early in life. These findings imply infant is wired for speech.

(2) Critical Period Hypothesis - Preschool children seem to have easiest time acquiring language. Believed to be between 2 and puberty. Support: Children who suffer brain damage before puberty usually recover their lost language functioning without special help. If damage after puberty you need extensive therapy just to get part of language back. Due to plasticity, right hemisphere takes over functioning.

(3) Linguistic Universals - If language is maturational all children should proceed through the same stages in acquiring their first language. This appears to be the case. Children with Down's Syndrome also proceed through the same stages but at a slower pace.

4. Problems with Nativist Approach

(1) Critical period is not the only time you can learn language. Studies show that with intensive training, adults do learn a foreign language at about same rate as children. Also, there have been cases of neglected children who had very little language when discovered after puberty but they learned language very rapidly.

(2) LAD was mainly developed to explain things researchers couldn't explain. There is no evidence to show it does exist.

5. Conclusion - Just about everyone agrees that there is a biological component but a strict nativist position is incomplete.

Interactionist Perspective

1. Interactionist Viewpoint - Both biological factors and the linguistic environment combine to influence language development. What is innate is not any specialized linguistic capacity but, rather, a nervous system that predisposes children to develop similar ideas at about the same time. Focus is on cognitive development.

Definition of Terms

1. Phoneme - The smallest unit of sound that distinguishes one utterance from another by signalling a difference in meaning. Languages differ in the number and kinds of phonemes they contain. English has about 45.

2. Morpheme - The smallest unit of a language that by itself has a recognizable meaning. It is a word or part of word. For example, word is one morpheme because it cannot be broken down. Words; however, is two morphemes (word and s).

3. Syntax - Consists of the rules by which words and morphemes are combined to form larger units such as clauses and sentences. Standard word order in English is subject-verb-object (She painted the picture). Other languages may be different.

4. Semantics - The study of meanings, how the sounds of language are related to the real world. Example: As adults if I said the sound "dog" you probably have an identifiable object of which you all agree. Children may not. Dog could mean any 4 legged animals for example.

LANGUAGE DEVELOPMENT

1. Infant's Reaction to Language - By 3 days of age an infant already recognizes his mother's voice and prefers it to other female voices. Young infants will suck harder to hear recorded speech than instrumental music. They therefore can discriminate speech very early in life.

2. Vocalizations - Infants are capable of vocalizing at birth and utter a number of speech-like sounds before they utter their first word. These vocal abilities develop in a step-like fashion over the first 10-12 months.

(1) Crying - Neonates are capable of producing at least 3 kinds of cries - a hunger cry, a mad cry, and a pain cry. During third week of life, many neonates develop a fake cry, they are not in discomfort or distress. One function is attention but it is also believed that it calms infant.

(2) Cooing - At about 3-5 weeks infants start to produce new sounds that are associated with pleasure. Vowel-like sound such as "oooh" or "aaah". Come when baby is awake, alert, dry, and contented.

(3) Babbling - At 3-4 months they add consonant sounds and by beginning of first year they are making consonant/vowel combinations such as "ka" and "ga." Soon the infant begins to repeat these combination "baba" or "kaka" that may sound like meaningful speech and this is babbling. Sounds that are made are very similar across cultures. Babbling continues until about 12-18 months. Deaf kids babble about same time as normal children but stop at about 8-9 months. This would suggest a maturational component in onset but environmental stimulation for it to continue. It appears its major function is to tune the speech muscles, to gain motor control over speech muscles to get ready for forthcoming verbalizations.

3. First Words - First word usually appears by end of first year. At first vocabulary is limited to one or two words that may be understandable only to family like "ba" for ball or "awa" for water. At first vocabulary increases very slowly, one word at a time, it usually takes about 3-4 months before vocabulary is 10 words. Once they reach 10 words; however, the pace quickens. By 19-20 months working vocabulary of about 50 words and by 24 months vocabulary of about 186 words. Most of their first 50 to 100 words are objects instead of actions.

4. Telegraphic Speech - At about 18-30 months children combine words into simple sentences, starting with 2-word utterances. Called telegraphic because they resemble the abbreviated language of a telegram. They consist solely of content words and omit less meaningful parts of speech like "Where ball?"

5. Preschool Period - Ages 2 1/2 to 5. They begin to produce lengthier sentences but important part is increase in complexity of sentences.

6. Refinement of Language - Although most of the English language system has been acquired by age 5, certain forms of syntax pose a problem for several years. They also learn more irregular verbs and nouns, become better in use of personal pronouns, etc.

PSYCHOLOGICAL TESTING

NORM- REFERENCED MEASUREMENT

1. Definition - In norm-referenced testing an examinee's performance is compared with the performance of a specific group. A raw score alone tells us nothing. What does saying someone got 17 out of 20 right mean. This allows us to compare a person's score with the typical performance of a group, usually one representative of the population of the country. Then we know how someone is performing compared to other kids their age.
2. Representativeness - This is extent norm group is characteristic of the population. Should match as closely as possible demographics of population. Usually use age, grade level, gender, geographic region, ethnicity, race, and SES.
3. Size - Should be large enough to insure stability of test scores and inclusion of all groups that are represented in population. Should include at least 100 subjects at each age and grade level.
4. Relevance - What is the norm group you are selecting? National norms are usually used in IQ tests. For educational achievement, local norms may be more appropriate. May want to compare only against their race or ethnic background. Depends on what you want to use the results for.

DERIVED SCORES

1. Age and Grade Equivalent Scores - This is arrived at by determining the average performance of children at an age or grade. You therefore find out at what age or grade the average child got the score of the examinee and that is their age or grade score. It does not mean that a child is functioning at that level. Grade scores should never be used. They are too misleading.
2. Ratio IQ - This is the first way that IQ's were calculated. It is $MA/CA \times 100$. This generally is not used anymore and if you see a test that uses this procedure you should question its utility.
3. Percentile Ranks - Allows us to determine an individual's position relative to the standardization sample. It is the point in the distribution at or below which that percentage of individuals fall. They are easy for people to understand, especially parents.
4. Standard Scores - The deviation IQ is a version of a standard score. These are raw scores that have been transformed to have a given mean and standard deviation. They express how far an individual's score lies from the mean in terms of the standard deviation.

RELIABILITY

1. Definition - Reliability refers to the stability of measurement. That is, it is reproducible and stable. It is expressed by a reliability (correlation) coefficient (range 0 to 1.0). A test should not be trusted if its reliability is low. Generally a reliability of .80 or higher is considered acceptable.
2. Test-Retest Reliability - An index of stability. You administer the same test to the same people at two different occasions, usually a short period of time (2 weeks to a month). The correlation represents the extent to which the score is stable over time.

3. Internal Consistency Reliability - Also called split-half reliability. Divide a test into two equivalent halves (usually odd versus even) and then obtain correlation.

VALIDITY

1. Definition - Generally it refers to does the test measure what it is supposed to measure? How meaningful is your result? Tests can be used for different purposes and therefore the validity of a test can vary depending on its use. IQ tests are a valid predictor of a child's success in school. It is not a valid predictor of your ability to be a janitor. Tests should be used only for those purposes for which its validity has been established. A test must be reliable to be valid but it can be reliable without being valid. Reliability is necessary but not sufficient.

2. Content Validity - Do the items on the test cover the domain that the test purports to measure. If you are measuring intelligence, does test cover all aspects of intelligence. It should not be confused with Face Validity which refers to what it appears to measure. It does not necessarily mean it has content validity.

3. Criterion-Related Validity - Refers to relationship between test score and some type of criterion or outcome such as a rating or other test score. Two types: (a) Concurrent Validity - How it relates to some other currently available measure. If came up with a new IQ test would see how it relates to standard in field, the WISC-R. (b) Predictive Validity - Refers to correlation between test score and performance on relative criterion after a period of time has passed. Is the test a predictor of future performance. Is IQ a predictor of future school performance?

4. Construct Validity - How does the test relate to the theoretical constructs that the test purports to measure. For example, does an IQ test measure the construct of intelligence the way the author designed. Usually done through factor analysis. WISC-R, does it break down to verbal and performance sections.

DEFINITION OF INTELLIGENCE

1. Binet (1916) - "...judgement, otherwise called good sense, practical sense, initiative, the faculty of adapting one's self to circumstances. To judge well, to comprehend well, to reason well, these are the essential activities of intelligence."

2. Terman (1923) - The ability to carry on abstract thinking.

3. Stoddard (1943) - "...the ability to undertake activities that are characterized by (1) difficulty, (2) complexity, (3) abstractness, (4) economy, (5) adaptedness to goal, (6) social value, and (7) the emergence of originals, and to maintain such activities under conditions that demand a concentration of energy and a resistance to emotional forces."

4. Freeman (1955) - "...adjustment or adaptation of the individual to his total environment, or limited aspects thereof....the capacity to reorganize one's behavior patterns so as to act more effectively and more appropriately in novel situations...the ability to learn...the extent to which a person is educable...the ability to carry on abstract thinking...the effective use of concepts and symbols in dealing with a problem to be solved..."

5. Wechsler (1958) - The aggregate or global capacity of the individual to act purposefully, to think rationally, and to deal effectively with his environment."

6. Das (1973) - "...the ability to plan and structure one's behavior with an end in view."

7. Humphreys (1979) - "....the resultant of the process of acquiring, storing in memory, retrieving, combining, comparing, and using in new contexts information and conceptual skills."

8. Conclusion - I don't care if you know all the different theories I just gave or the ones given in the book. Just remember that there are two major divisions in theories. Those that view intelligence as a unitary construct and those that see it is having many components.

HISTORY OF INTELLIGENCE TESTING

1. Roots - Interest in intelligence and intelligence testing began in earnest during the latter part of the 19th century as psychology began to emerge as a discipline of its own.

2. Galton (1869) - Some regard him as the father of the testing movement. He was concerned with the study of individual differences and began studying mental inheritance.

3. Binet (1905) - With Simon developed the Binet-Simon Scale, the first test of intelligence to have valuable practical application.

4. Terman (1916) - Published the Stanford Revision and Extension of the Binet-Simon Intelligence Scale.

5. Wechsler - Published the Wechsler-Bellevue Intelligence Scale in 1939. Since that time the Wechsler scales (WAIS, WISC, WPPSI) have become the most frequently used tests

INTELLIGENCE CLASSIFICATIONS

1. Profound Retardation - IQ < 20

2. Severe Retardation - IQ = 20-39

3. Moderate Retardation - IQ = 40-54

4. Mild Retardation - IQ = 55-69

5. Borderline Retardation - IQ = 70-79

6. Dull Normal - IQ = 80-89

7. Average - IQ = 90-109

8. Bright Normal - IQ = 110-119

9. Superior - IQ = 120-129

10. Very Superior - IQ = 130+

INTELLIGENCE USED TODAY

1. WISC-3 - Appropriate for children age 6 through 17. It is the most commonly used test today. Provides you with a Verbal IQ (measure of verbal comprehension), Performance IQ (perceptual organizational skills), and Full Scale IQ.

2. WAIS-R - Appropriate for individuals ages 16 through 74. Most commonly used adult test. Provides you with same scores as the WISC-R.

3. WPPSI-R - Appropriate for children 3 to 7 years of age. Also gives VIQ, PIQ, and FSIQ.

4. Stanford-Binet - Revision just came out a little more than a year ago. Appropriate for ages 2 to adult. Provide measures of verbal reasoning, abstract/visual reasoning, quantitative reasoning, short-term memory, and a composite IQ.

STABILITY OF IQ

1. Infant Intelligence Tests - Tests of IQ during first year of life do not accurately predict IQ performance later in childhood although they are useful in identifying motor problems or extreme IQ deficiencies. These tests are made up primarily of sensorimotor tasks.
2. Later Testing - From about age 8 on the tests become very good predictors of adult intelligence although there is still a lot of variability.
3. Prediction - IQ tests were generally developed to predict how well someone will do in school. They do this very well.

GENETIC HYPOTHESIS

1. Controversial - This is a very controversial area that explains differences between groups (black and white) in intelligence are inherited.
2. Arthur Jensen - The strongest proponent of this position. He believes that lower-class and black children are not as good as middle class and white children in abstract reasoning and problem solving (higher level intellectual abilities).
3. Study - He was initially not looking to test a genetic hypothesis but rather he was looking for a cumulative deficit. This would be an increase in decrement of IQ with age as a result of environmental deprivation.
4. California Study - First attempt to find this. Looked at children 5-10 years. Found age decrement for black children on verbal IQ. Also found that overall, blacks scored about 1 standard deviation below whites.
5. Georgia Study - To further test the hypothesis of a cumulative deficit he looked to find the worst environment he could. Found it in rural Georgia. Tested children in grades K-12 in an integrated school (1/2 white, 1/2 black). Blacks were very low SES, whites were low to low middle SES. Findings: (a) IQ - White = 102 Black = 71 (b) significant age decrement in blacks but not whites in both verbal and performance IQ. He concluded difference due to heredity.
6. Results and Types of Intelligence - Level I intelligence is rote memory. Mostly short-term memory tasks like digit span. Level II intelligence involves more complex cognitive processing and problem solving skills. Jensen found the Black-White difference on Level II only. He also found Blacks to do the worst on the least culturally loaded tests and best on vocabulary and information which are the most culturally loaded. Other studies have not confirmed this however.
6. Criticisms - If true would expect mixed-race persons to be lower than whites. When compared to white children of similar backgrounds there was no difference in IQ. In general there is no evidence to support this or deny it. That is why it continues to be perpetuated.

HEREDITY- ENVIRONMENT ISSUES

1. Studies in Families - (a) In general, the more the genetic similarity, the higher the correlation. (b) It does however support the role of both heredity and environment..
2. Study With Adopted Children - Look at scores of children adopted before 6 months into middle SES homes. Group I: Mother had IQ<75. Group II: Biological fathers occupation was at very low level. Group III: Biological parents both I and II. Results: Tested children at 8 years. I, IQ=105; II, IQ=110; III, IQ=104.

3. Conclusion - Heredity may set limits but the rest is determined by how much stimulation you get in your environment.

CHILD DEVELOPMENT

Definition of Development

1. Definition of Development - the process by which organisms grow and change over the course of their lives. When does it begin? From time sperm fertilizes the egg. When does it end? Death.

2. Areas of Development

Physical

Perceptual

Cognitive

Language

Socio-Emotional

3. Nature - Nurture - This is the main issue in the study of development.

(1) Nature: Some people believe that much of human behavior is guided by genetics, physiological maturation, and neurological functioning. There is a biological program that determines the course of your life. Universals in behavior (e.g., walking, talking, etc.) can be easily explained from this perspective. Individual differences are also believed to be genetic (Intelligence - poor kids do worse on IQ tests because of genetic reasons and poor prenatal care but they are born that way and no matter what you do you can't change them).

(2) Nurture: Emphasizes the physical and social environment on development. You respond to people and objects in your environment and that are how you change. IQ differences are due to cognitive stimulation and opportunities to learn, anyone can do well with the right environment.

(3) Integration: Both extremes have weaknesses. Most psychologists agree that both biological and environmental variable play a role. Some biological role more important (walking), some experience (learning to read), some both (talking, intelligence, etc.).

4. Stability - Is behavior stable over time? If you have an aggressive 3 year old, does that mean you will have an aggressive adult? No clear answer. Some people stress the individual changes as they grow, others stress the stability over time. In general, behavior gets more stable as you get older. Example: IQ measured at age 2 is not that good a predictor but IQ measured at 10 is. Also, some behaviors are more stable than others. Aggression is fairly stable. Does that mean it is not changeable? NO. Could be environment that remains same.

PRENATAL DEVELOPMENT

1. Stages of Prenatal Development - Zygote (first 2 weeks of life). Begins at fertilization and extends until zygote (fertilized egg) becomes implanted in the wall of the uterus. Embryo (3rd to 8th prenatal week). Differentiation of most important organs and physiological systems occurs at this time and by end of this period the embryo is recognizable as a tiny human. Rapid growth, the embryo increases 2 million percent in size but because of this rapid growth the organism is most vulnerable to adverse environmental effects. Fetus (From beginning of 3rd month until birth). Little further

differentiation of organs occurs, however, muscle development is rapid, closure of the palate occurs, and genitalia differentiation continues. Central Nervous System also develops rapidly. At about 4 months you get movement of fetus. At 5 months reflexes such as sucking, swallowing, and hiccoughing appear. At about 6 months eyes develop and can open and close eyes. Infant still can't survive outside mother yet. 28 weeks usually thought of as age of viability, there are exceptions however.

PRENATAL INFLUENCES ON DEVELOPMENT

1. Teratogen - Any external agent that can harm a developing embryo or fetus. Can include disease, blood disorders, diet, radiation, hormones, temperature, or oxygen level.
2. Developmental Stage - The effect of the teratogen varies with prenatal age. The system that is developing at the time of the teratogen is most vulnerable. Most vulnerable period for brain is 15-25 days, for eyes 24-40 days, heart 20-40 days, legs 24-36 days. During fetal stage the effects are limited to those systems that are still developing mainly the cerebellum, palate, cardiovascular system, nervous system, or genitalia.
3. Different Systems - Different teratogens influence specific developmental processes. Rubella effects heart, eyes, and brain. Thalidomide results in malformations of limbs. Therefore, the most dangerous times for these things is when those systems are developing.

PHYSICAL DEVELOPMENT

1. Norms - These are approximate ages when important attributes and skills are acquired in average child.
2. First 3 Months - Sleeps most of the time. Small but steady changes in perceptions of world (what he sees and hears).
3. About 3 Months - More visual involvement. Exploring and interacting with world with eyes.
4. 4-7 Months - Improved eye-hand coordination. At 5 months get first purposeful, eye guided reaching for objects.
5. 6 Months - Rolls over completely: back to stomach and stomach to back.
6. 7 Months - Sits up without help.
7. 8-9 Months - Begins to crawl. First done on belly with feet dragging.
8. First Half of Second Year - Usually takes first steps.
9. 18 Months - Infant can run but it is very clumsy.
10. 2 Years - Walks and runs fairly well.
11. Variability - These only tell us what average, not what is normal or abnormal is. If someone is faster or slower it doesn't mean they are smarter or will develop at that rate min everything.

METHODS OF STUDYING DEVELOPMENT

1. Cohort Effects - Your cohort are people who were born around the same time you were. Cultural effects change from period to period and from place to place - it must effect development.

2. Cross-Sectional Design - This takes different age groups at the same point in time in order to make conclusions as to how development proceeds. It gives idea of change over time and clues to the mechanisms of development.

5. Longitudinal Design - This takes a group of children the same age and follows them over a period of time to see how development progresses. Can range in time, from as little as one to two years to a lifetime.

COGNITIVE DEVELOPMENT

1. Piaget - Theory goes from birth to adulthood. His theories are the foundation, frame of reference, for all cognitive work. Cognitive development occurs through 4 stages and each one is qualitatively different from the other. Children learn by actively manipulating their environment. This is done through assimilation and accommodation. Assimilation is using your existing mental structure to understand new information. Accommodation is changing your cognitive structures to fit new information.

2. Stages of Development

(A) Sensorimotor Period (Birth to 2 years) - The infant becomes more externally motivated. In the beginning, everything is centered on the body, with no distinction between external and internal world. By the end of this period, the child has developed a basic capacity for symbolic thinking; representing events that are not immediately present. Prior to this, everything is done by trial and error.

(B) Preoperational Period (2-7 years) - Children become increasingly proficient at constructing and using mental symbols to think about objects, situations, and events. Despite these advances there are limitations as they have not yet acquired all schemata necessary for logical thinking. Now have symbolic thought. Main deficit in reasoning is egocentrism, the inability to distinguish their own perspective from that of others.

(C) Concrete Operational Period (7-11 years) - Cognitive abilities are increasing dramatically. Develop ability to internalize actions and to reason in an elementary fashion. They are still tied to concrete objects in the immediate world. They can solve problems only if the objects to solve them are physically present.

(D) Formal Operations Period (12 and up) - Can now reason quite logically about abstract ideas that have no basis in reality. Thought is much more flexible and abstract.

3. Moral Development - The process by which children acquire society's standards of right and wrong. Kohlberg built on Piaget's work and studied morality by focusing on moral reasoning, the thinking the child displays in deciding if something is right or wrong. Basic assumption is that moral development depends on cognitive development. Development proceeds through a series of stages.

4. Levels and Stages - There are 3 moral levels and 6 stages.

Level 1: Pre-conventional Morality

(1) Stage 1: Punishment and Obedience Orientation - Child determines goodness or badness or an act on basis of its consequences. Obeys rules to avoid punishment. Doesn't understand rules, if child can get away with it, it is not considered bad. The seriousness of violation depends on how much punishment you get.

(2) Stage 2: Naive Hedonism, or Instrumental Orientation - Conforms to rules in order to gain rewards or satisfy personal needs. What is right will benefit you in long run. Philosophy is "You scratch my back, I'll scratch yours."

Level 2: Conventional Morality

(3) Stage 3: "Good Boy" or "Good Girl" Orientation - Moral behavior is what pleases, helps, or is approved by others. Prime objective is to be thought of as a nice person.

(4) Stage 4: Authority and Social-Order-Maintaining Morality - One accepts and conforms to rules and conventions in order to avoid censure by legitimate authorities. Reason for obeying rules is not fear of punishment but belief law and order worth preserving.

Level 3: Post-conventional Morality

(5) Stage 5: Morality of Contract, Individual Rights, and Democratically Accepted Law - Flexibility in moral judgements. Moral actions are those that express the will of the majority or maximize social welfare.. Rules must be impartial and arrived at democratically. Rules can be modified if majority wants them to. This is morality of U.S. Constitution.

(6) Stage 6: Mortality of Individual Principles of Conscience - Highest stage of moral reasoning. Defines right and wrong on basis of self-chosen ethical principles of one's own conscience. Not concrete like 10 Commandments but are more abstract and can be applied to all situations. Deviations leads to guilt and self-condemnation.

SEX DIFFERENCES

Answer true or false

1. At the age of 2 1/2, boys and girls typically think that their own sex is superior. True. Chauvinism begins early.
2. Three year olds report that women are more likely to cry than men. True. Knowledge of stereotypical sex roles is forming rapidly by this age.
3. Girls are superior to boys in verbal abilities. True. Girls speak somewhat earlier than boys, have crisper pronunciation, and encounter fewer reading problems.
4. Fathers are more likely than mothers to help children with their math homework. True. Math is generally seen as part of the male domain in our culture.
5. Girls are more likely than boys to have math anxiety. True. It is difficult to convince even highly capable girls to take math courses.
6. Girls are more likely to play with "boys' toys" than boys are to play with "girls' toys." True. Sexism leads boys to view girls activities as inferior.
7. Girls do not like to play with guns. True. Nor, generally speaking, do they like boxing or other rough and tumble activities.
8. Girls like to go to school but boys do not. False. Boys and girls both generally like school.
9. Boys are more aggressive than girls. True.
10. Boys are more likely than girls to curse. True. Boys are also more likely to issue commands, while girls are more likely to make polite requests.
11. Women who compete with men in the business world are suffering from penis-envy they experienced as little girls. False. This psychoanalytic notion is based on ignorance and prejudice.
12. A 2 1/2 year old may know that he is a boy, but still thinks he can grow up to be a mommy. True. Children do not recognize that they will maintain their genders until the concept of gender stability emerges at about the ages of 4 or 5.
13. Adolescent girls who show a number of "masculine" traits have higher self-esteem than girls who thoroughly adopt the traditional feminine sex role. True.

14. Females are the weaker sex as far as physical vulnerability is concerned. False. Girls develop physically and neurologically quicker than boys, have a lower rate of infant mortality, and are less vulnerable to disease and hereditary abnormalities.
15. Almost all students who are labelled as exceptionally talented in mathematics by the junior high school level are male. True. Boys excel in mathematics beginning at about age 12.
16. Girls are more compliant to the demands of parents and other adults as early as age 2. True.
17. Boys are more likely to have school problems, reading disabilities, speech defects, and emotional problems. True.
18. Girls are more social than boys. False. Boys and girls spend about as much time with others and are equally responsive to others.
19. Girls are more suggestible than boys. False. Girls are not more likely to conform to standards of a peer group or to imitate the responses of others.
20. Boys are more responsive to visual stimuli and girls to auditory stimuli. False. No sex differences.

MOTIVATION & EMOTION

DEFINITIONS

1. Motivation - Internal processes that serve to activate, guide, and maintain our behavior. In laymen terms you could say it is how much you want to do something.
2. Emotion - These are complex reactions we have and consist of (1) physiological component (BP, HR, etc.), (2) subjective cognitive state (joy, anger, etc.), and (3) expressive reactions (facial expressions, posture, etc.).
3. Related Concepts - Emotions and MOTIVATION are closely related. For example, if your girlfriend or boyfriend just broke up with you, you may feel depressed. Your motivation to do your homework may therefore be decreased.

THEORIES OF MOTIVATION

1. Instinct Theory - This is the most basic theory that proposes that we are motivated to act in certain ways because we are born with an innate predisposition to act that way. We are aggressive because we are born with an aggressive instinct. An instinct is believed universal in a species. Not everyone is aggressive however. That is a flaw in this theory.
2. Drive Theory - Drive theory is somewhat similar to instinct theory in that it believes there are innate drives to satisfy basic biological needs. When a need is not met it creates an unpleasant state of arousal which we are motivated to decrease. To decrease this state you act in a way to fulfil the need. This too has flaws. For example, how do you explain dieting or eating disorders which leads only to an increase in a basic drive (hunger)?
3. Arousal Theory - This theory proposes that we seek to obtain a level of arousal that is optimal for us in any given situation. Usually this is some moderate level of arousal, you don't want to be too relaxed or too excited.
4. Expectancy Theory - This is based on your cognitive concept of expectancy, your belief that your behavior will yield a certain outcome. When your expectation is for success or something pleasurable, your motivation is high. When the expectation is for

failure or something unpleasurable, motivation decreases. For example, say you were considering asking someone for a date. If you expect them to say yes, your motivation to ask them would be high. If you expect them to say no, your motivation would be low.

5. Maslow's Need Hierarchy - This theory is popular, especially in business. He proposes that individuals have needs that exist in a hierarchy and you are not motivated to satisfy higher level needs unless your lower level needs have been satisfied. At the base of the hierarchy are physiological needs (food, water, oxygen, sleep). Next comes safety needs, our need for feeling safe and secure. Then comes social needs which are our need to be loved and accepted. These three needs are referred to as deficiency needs because these are basic to everyone and you can't think about personal growth unless these needs are met. The higher level growth needs are esteem needs, the need to develop self-respect and success; and self-actualization needs, which involves self-fulfilment and being all you can be. He believed that only exceptional people like Gandhi and Martin Luther King ever reach this top level.

HUNGER

1. Hunger Motivation - For most people, hunger is a potent motivator. If you are hungry, you will be highly motivated to get food. While not always the case in our modern society, it has been studied heavily.

2. How Do We Know We Are Hungry? - Our body needs nourishment in order to survive. We have certain receptors in our body that signal to us if we need nourishment or are full. Unfortunately it is much more complex than simple biology. There are many behavioural and cultural cues that may make us eat even when our body tells us we are not hungry.

3. Regulation of Body Weight - Some people seem to be able to eat anything and never gain an ounce and others look at food and gain weight. Why? One explanation is basal metabolic rate. Some people require more calories to maintain than weight than others. It is not that simple, however. Our culture rewards thinness so there are strong external cues not to eat too much. Stress also can influence eating, it reduces stress in some people. People may also be conditioned to other external cues such as time. Some research has shown that obese people are more controlled by external cues such as time of day or the presence of eating cues than others. I never met a donut I didn't like.

4. Eating Disorders - Anorexia and bulimia have become major problems, especially among females, in our society. Anorexia is an extreme fear of being overweight characterized by a refusal to eat even enough to stay healthy and a distortion of body image. 95% of anorexics are female and 1 in 250 girls 12 to 18 have this disorder. Bulimia is repetitive bingeing and purging. It is more common with 5% of women having some history of it.

SEXUAL MOTIVATION

1. Potent Motivator - There is no question that sex is a very potent motivator. Everywhere you look you see sex being advertised from clothes, to food, to liquor, to cigarettes.

2. Hormones - Especially in lower animals, hormones play a large role in sexual behavior. Without hormones, many animals show no interest in sex. They play a role in human sexual arousal also but cognitive factors also play a very important role.

3. Human Sexual Response - Sexual response can be divided into four phases. The Excitement Phase consists of physiological changes. The heart beats faster, respiration increases, and blood flows to the genitals which results in the penis becoming erect or swelling in the clitoris and vaginal lubrication. If sexual stimulation continues you enter the Plateau Phase. The penis becomes even more erect, the entrance to the vagina contracts and BP and HR continue to rise. After continues stimulation the Orgasm (Organismic Phase) occurs. This consists of contraction of muscles around the genitals and a feeling of pleasure. This is when the male ejaculates. Up to this point males and females react fairly similarly. Genders differ in the Resolution Phase however. Males show a reduction in sexual and physiological arousal and they enter a refractory period when then are unable to have another orgasm. Females, however, can experience continued orgasms if stimulation continues.

4. Cultural Factors - While this is the general pattern of physiological response there is a great deal of variability in responding and in the pleasure of the response due to cultural factors. for example, if you are brought up to believe sex is bad you probably will not be as easily stimulated and the sex act may not be as pleasurable.

5. Sexual Orientation - Sexual orientation is your preference for sexual relations with the opposite sex, your own sex, or both. About 4% of males and 2-3 % of females are exclusively homosexual. Another 2-3% of each sex consider themselves bisexual. Homosexuality and Bisexuality have been present in all cultures throughout history, however, the origin of homosexuality is still not clear. There is no clear evidence that homosexuality is a choice however. It has nothing to do with a dominant mother and a weak father and there is no evidence to support environmental factors. There is incomplete support for a biological basis but this is still inconclusive.

THEORIES OF EMOTION

1. Cannon-Bard and James-Lange Theories - These are similar theories. The Cannon-Bard Theory proposes that emotion provoking events simultaneously produce subjective reactions (emotions) and physiological arousal. The James-Lange Theory proposes that first you have the emotion producing event, then you experience a physiological reaction, and then you label this reaction with your subjective feeling of emotion.

2. Schachter & Singer's Two-Factor Theory - Emotion producing events cause internal arousal, we then attend to the external environment to help us label our arousal, based on our interpretation of the stimuli in our environment we label our arousal which determines the emotion we perceive. it is called two-factor because it includes a physiological and cognitive factor. Example, did you ever see a child fall and then look around to see if they should cry. If the mother shows concern they cry, if the mother doesn't appear concerned they get up and start running again.

OUTWARD SIGNS OF EMOTION

1. Nonverbal Cues - In order to have greater control over our world we read into the behavior of others what emotions they are experiencing. It helps us decide how we should respond to them. We do this by reading various nonverbal cues that they give us.

2. Facial Expressions - Emotions are often expressed by facial expression and how another person looks facially gives us an understanding of how they are feeling. There are six basic emotions registered by the face: happiness, sadness, surprise, fear, anger, and disgust. Facial expressions appear to be fairly universal, being similar in all cultures.

3. Eye Contact - We place a great deal of importance to eye contact. If someone doesn't make eye contact with you when talking with you it could be a sign of shyness, that they are not interested or even deception. In any case, we generally have a negative reaction to people who fail to make eye contact.

4. Body Language - Body posture and movement of body parts can frequently give us cues to the emotion the other person is experiencing. Fidgeting a lot is usually a sign of anxiety. Crossing your arms across your chest could be a sign of defensiveness. Leaning forward with open arms could be a sign of openness.

SOCIAL PSYCHOLOGY

SOCIAL PSYCHOLOGY DEFINED

- **Social psychology** is a science that studies the influences of our situations, paying special attention to how we view and affect one another. Also how people think, influences, and relate to one another.

SOCIAL PSYCHOLOGY BIG IDEAS

- According to **Myers (2010)**, some of the big ideas in social psychology:
 - we construct our social reality
 - social intuitions are powerful and sometimes perilous
 - our behavior is shaped by our attitudes and social influences
 - dispositions shape our behavior
 - feelings and actions toward people can be negative and positive
 - biological behavior influences our social behavior
- **Attributions** affect the way we judge others motives. One distinction we make is between personal and situational attributions.
- When we make a **personal attribution**, we believe the cause of someone's behavior is a personal characteristic of theirs.
- **Situational attributions** are made to environmental causes.
- **Attitudes** are most predictive of behavior when situational factors are weak, and we are all guilty of being hypocrites because of our attitudes and actions.
- When we are in social situations, we often find ourselves changing our opinion, at least explicitly. We tend to conform to numerous beliefs even though we do not really appreciate them. We may also change our opinions from group to group, to suit the majority opinion.
- There are others who often found themselves being very judgmental towards others; their values and beliefs and their worst habit is criticizing because they felt their thoughts deemed truer.

THE SELF IN THE SOCIAL WORLD

- **Spotlight effect** is the belief that others are paying more attention to one's appearance and behavior than they really are.
 -Example: They like Piolo Pascual because of his good- looking appearance.
- **Illusion of transparency** is the illusion that our concealed emotions leak out and can easily be read by others.
 -Example: If a student had an assignment due in class and did not prepare as well as they should have, the student may start to panic and think that simply because they did not prepare well, the teacher will know and call on them for answers.
- **4 aspects of the self:**
 - self-concept (who am i)
 - self-knowledge (explain self)
 - self-esteem (sense of self-worth)
 - social self (social roles)
- **Social schema** beliefs about self that organize the processing of self-relevant information.
 - Types: role schema, events schema, and self- schema
 - Example: **role schema:** If Joe is employed as a physician; he displays his professional role in the hospital but likely behaves differently while with his family on vacation. As observers, our expectations about how he should act differ depending on the social situation.
 - Example: **events schema:** If Sally tends to be shy and quiet by nature, then that is how the people who know her expect Sally to behave at a party full of strangers. If she suddenly acts very outgoing at a party, that'd be a big surprise to her friends!
 - Example: **self- schema:** If Mario believes that he is an outgoing and people like talking to him at parties, then he will feel obligated to modify his behavior to fit this self- schema.
- **Collectivism** is giving priority to the goals of one's groups and defining one's identity accordingly.
 - Example: In politics, such as a representative democracy. In this system, citizens vote for their leader, and after the votes are counted, you are expected to accept this leader, even if it wasn't the leader you voted for.
 - Example: In socialist economic system, joint ownership of farms was a common practice in the former Soviet Union. Farmers would own land collectively and benefit from pooling their labor and resources. In more extreme examples of collectivism, private property is eliminated entirely.
- **Interdependent self** is construing one's identity in relation to others.
 - Examples: I am Nancy's best friend.
 - I'm the eldest son in my family.
- **Planning fallacy** is the tendency to under-estimate how long it will take to complete a task.
 - Example: In planning a business change project, the CEO assumes that everyone will cooperate and attend all meetings and not object to any of the

changes. Of course this does not happen as expected and the subsequent issues make the project drag on for three times as long as he expected.

- **Impact bias** is overestimating the enduring impact of emotion-causing events.
 - Examples:* People overestimated how unhappy they would be two months after the dissolution of a romantic relationship.
 - : Untenured college professors overestimated how unhappy they would be five years after being denied tenure.
- **Immune neglect** is the human tendency to underestimate the speed and strength of the "psychological immune system," which enables emotional recovery and resilience after bad things happen.
 - Example:* When asked to predict how they will feel two months after a breakup, people often expect to feel devastated. But when you compare them with those who have already gone through a breakup, you find that the effects tend to be milder and wear off much more quickly than was predicted.
- **Dual attitudes** is differing implicit (automatic) and explicit (consciously controlled) attitudes toward the same object.
 - Example:* Mary is convinced that she no longer loves his husband Cesar and should end her relationship with him. When her housekeeper rushes into the room with the news that Cesar has left her, however, she realizes that she still love him.
- **Self-esteem** is a person's overall self- evaluation or sense of self-worth.
 - narcism and self-esteem interact to influence aggression.
 - Example:* Martin believes that he is a bad person, and feels bad about himself in general.
- **Self-efficacy** is a sense that one is competent and effective.
 - Example:* "I am confident that I will be able to grasp the main ideas in this chapter." "If I read this section carefully, I will be able to explain what self-efficacy is." "I know that I am capable of scoring an A on a test of the material in this book." If you agree with these statements, you have high self-efficacy for mastering educational psychology; if you mainly disagree with these statements, your self-efficacy could be classified as low.
- **Locus of control** is the extent to which people perceive outcomes as internally controllable by their own efforts or as externally controlled by change or outside forces.
 - Example:* Students with an "internal locus of control" generally believe that their success or failure is a result of the effort and hard work they invest in their education. Students with an "external locus of control" generally believe that their successes or failures result from external factors beyond their control, such as luck, fate, circumstance, injustice, bias, or teachers who are unfair, prejudiced, or unskilled. Moreover, students with an internal locus of control might blame poor grades on their failure to study, whereas students with an external locus of control may blame an unfair teacher or test for their poor performance.
- **Learned helplessness** is the sense of hopelessness and resignation learned when a human or animal perceives no control over repeated bad events.

-The mantra of the person who suffers from Learned Helplessness is: "What's the point in trying?"

-*Example:* A woman fears her father's angry outbursts. Sometimes he becomes violent but usually his weapon of choice is a verbal torrent of criticism. The woman tries to avoid her father as much as possible, and tries to be quiet when he is present, hoping he will not notice her. She may begin to feel shame and avoid talking to others because she does not want them to know what is happening.

This situation is more complex because they are in unchosen relationship (in this example, with a parent) may have a history where they didn't have the same power of independence as an adult. However, when she tries to become invisible, a cycle of depression and shame has begun. This cycle may then be carried outside of the home and into other areas of her life, where she is not being abused. This may contribute to feelings of depression, isolation and helplessness. In reality, she can do nothing to control her parent. She can, however, work on her own safety and protection.

- **Self-serving bias** is explanation for positive and negative events.

-The tendency to perceive oneself favourably.

-*Example:* A student who attributes earning a good grade on an exam to their own intelligence and preparation but attributes earning a poor grade to the teacher's poor teaching ability or unfair test questions is exhibiting the self-serving bias.

- **Self-serving attributions** is a form of self-serving bias; the tendency to attribute positive outcomes to oneself and negative outcomes to other factors.

-false consensus effect the tendency to overestimate commonality of one's opinions and one's undesirable or unsuccessful behaviors.

-false uniqueness effect the tendency to underestimate the commonality of one's abilities and one's desirable or successful behaviors.

-*Examples:* If you notice that every time you tell a certain joke, people laugh, then you might attribute people's laughter to the joke, and you're likely to be correct.

: If you notice that every time you don't study for a test, you get a bad grade, then you will start to attribute bad grades to not studying.

- **Defensive pessimism** is the adaptive value of anticipating problems and harnessing one's anxiety to motivate effective action.

-*Example:* There is a possibility that too many questions will raise by my students in this topic that I am going to discuss in the class tomorrow, so I have to research more on this and read and analyze this topic many times!

- **Group-serving bias** is explaining away out-group members' positive behaviors; also attributing negative behavior to their dispositions (while excusing such behavior by one's own group).

-*Example:* REAA political party spokesperson said that they did not become successful in political arena because they are not lucky and don't have a special advantage compared to others. But lack of commitment of each member is the true reason.

- **Self-handicapping** is protecting one's self image with behaviors that create a handy excuse for later failure.
 - Examples:* If people predict they are going to perform poorly on tasks, they create obstacles, such as taking drugs and consuming alcohol, so that they feel that they have diverted the blame from themselves if they actually do fail.
 - In addition, another way that people self-handicap is by creating already-made excuses just in case they fail.
 - Example:* If a student feels that he is going to perform badly on a test, then he might make up an excuse for his potential failure, such as telling his friends that he does not feel well the morning of the test.
- **Self-presentation** is the act of expressing or behaving in a particular way in order to present oneself to a certain level of ideals.
 - Example:* A woman may interact with many people during the day and may make different impressions on each person. When she starts her day at her apartment, she chats with her roommates and cleans up after breakfast, thereby presenting the image of being a good friend and responsible roommate.
- **Self-monitoring** is being attuned to the way one presents oneself in social situations and adjusting one's performance to create the desired impression.
 - Example:* George, a high self-monitor, enjoys keeping the peace among his friends. Even when he disagrees with one of them, he remains very polite and rarely asserts his opinion.
- **James (1890) levels of the empirical self (in order of importance)**
 1. spiritual self (inner self- personal perceived abilities, attitude, emotions, etc.)
 2. social self (stats, background, role)
 3. material self (body and extended self)
 - most damage=no recognition
- **8 contemporary components of the self-concept:**
 1. interpersonal attributes (daughter)
 2. ascribed characteristics (female, 19)
 3. interests/activities (photography)
 4. self-determination (educational goals)
 5. existential aspects (friendly, kind)
 6. internalized beliefs (democrat, feminist)
 7. self-awareness (good person)
 8. social differentiation (idahoan, come from small, lower class family)
- **Few of the aspects that relate development of the social self -the roles we play:**
 - social identities
 - social comparison
 - success/failure experience
 - other people's judgments
 - self and culture

SOCIAL THINKING

- How do we think about one another?
- **Attribution Theory** pertains to idea that we give a casual explanation for someone's behavior.
 - We credit that behavior either to the situation or to the person's disposition.
 - Example:* Was my friend a jerk because she had a bad day or is just a bad person?
- **Fundamental Attribution Error** is the tendency to underestimate the impact of a situation and overestimate the impact of personal disposition.
 - Example:* How do you view your immediate superior? You probably attribute it to their personality rather than their profession/position.
- **Attribution at Work** (the picture explains this concept).
- **The effects of attribution:**
 - social effects
 - political effects
 - workplace effects
- **Attitude** is a belief or feeling that predisposes one to respond in a particular way to something.
- **Do our attitudes guide our actions?**
 - only if external pressure is minimal, we are aware of our attitudes, and the attitude is relevant to the behavior.
 - more often, our actions affect our attitudes.
- **Foot-in-the-door phenomenon** is tendency for people who have first agreed to a small request to comply later with a larger request.
 - Example:* If I give out an answer on a quiz, what happens next?
- **Door-in-face Phenomenon** is tendency for people who say no to a huge request, to comply with a smaller one.
 - Example:** If I ask my mom for the 1952 Topps Mantle card (\$15k) she will say? NO
 - : But she may let me buy a new play station game.
- **Zimbardo Prison Study.** Role playing affects attitudes.
 - Example:* What do you think happened when college students were made to take on the roles of prison guards and inmates?
- **Cognitive Dissonance Theory.** We do not like when we have either conflicting attitudes or when our attitudes do not match our actions. When they clash, we will change our attitude to create balance.

SOCIAL INFLUENCE

- The actions, reactions, and thoughts of an individual are influenced by other people or groups.
- Social influence may be represented by peer pressure, persuasion, marketing, sales, and conformity.
- **Sources of social influence:**

-taking a broad perspective, we can think of social influences at the three levels of analysis recognized by sociologists. In a nutshell, people are influenced in many different ways by:

1. **Social institutions:** Organized religions, political parties, and labor unions are social institutions that influence our attitudes, beliefs, values, and behavior
 2. **Interactions with other people:** The people we interact with, at home, at work, or at play.
 3. **Individual socialization:** Beginning in infancy, the process by which we are inducted into a culture or a society.
- Individuals are as unique as fingerprints. Nevertheless, the degree to which people are open to social influence depends on how they are socialized. The language we speak, the ideas we hold to be true, and all the ways we are likely to behave are products of socialization.
 - **Automated Behaviors.** According to Prof. Robert B. Cialdini (University of Arizona), our behaviors, ideas, and beliefs are automated. Put another way, an awful lot of what we do, what we say, and what we believe is a product of habit. We internalize all kinds of habitual, virtually automatic behavior that's similar to a reflex response to a physical stimulus.
 - Example:* If the doctor taps near your kneecap with a little rubber hammer, you'll exhibit a knee-jerk reaction. In a similar way, humans will react automatically, and often unconsciously, to social stimuli.
 - **Principle of Social Influence.** "Where all think alike, no one thinks very much"- Walter Lippman (one of the founders of Modern Public Relations).
 - **The Reciprocity Principle.** You scratch my back, and I'll scratch your back. That aphorism helps us think about the reciprocity principle. It's a principle found in every culture.
 - In fact, Richard Leakey, a much-respected paleoanthropologist, maintains that the reciprocity principle is an essential human trait that helps us sustain ideals of sharing and cooperation within social groups.
 - On the dark side, the rule of reciprocity can be used to manipulate behavior.
 - Examples:* Companies provide free samples to potential buyers. Often enough, that may encourage a person to buy Product X- under the social rule of reciprocity. Further, to the extent that reciprocity rules in market economics, people may be encouraged to assume more debt that they can manage.
 - : In social situations, you may feel you should invite Jake to your brother's birthday party, even though you don't really like him, simply because he invited you to his mom's retirement party.
 - **Commitment and Consistency.** Prof. Cialdini refers to the commitment and consistency principles as **hemoglobin** of the mind. Here are examples of both of these principles that illustrate his point. Let's look at these principles separately:
 - Commitment:** If your friend can get you take a stand on something like, say, supporting charter schools, you will have made a commitment. That may be the case even though you actually agreed to be a proponent of charter schools because you want this person to like you. Nevertheless, when you attend a debate on the charter school issue, you'll feel compelled to voice your approval of

charter schools over public schools - even if your friend isn't present at the debate.

Consistency: Think again of the about the example we just cited. In ordinary, day-to-day life, we are generally rewarded for being consistent. For example, if you are generally argue in favor of lower taxes, you'll feel uncomfortable arguing for higher taxes, even if you sense at some level that some common need, such as improved roads are, might justify higher taxes. To appear consistent to your peers and family, you'll argue you low- tax position come hell or high- water.

-Now, we can see commitment and consistency are often strongly related. Both are **hemoglobin** of the mind because they motivate foolish consistency and on the one hand and foolish commitment on the one hand.

SOCIAL RELATIONS

- How do we relate to others?
- **Prejudice** is an unjustifiable attitude towards a group of people.
 - Usually involves stereotyped beliefs (a generalized belief about a group of people).
- **Social Inequalities** is a principle reason behind prejudice.
 - In-group: "us"- people with whom one shares a common identity.
 - Out-group: "them"- those perceived as different as one's in-group.
 - In-group bias: the tendency to favor one's own group.
- **Scapegoat Theory.** The theory that prejudice provides an outlet for anger by providing someone to blame.
- **Why there is prejudice?**
 - Categorization
 - Vivid Cases (Availability Heuristic)
 - The Just-World Phenomenon
- **Aggression** is any physical or verbal behavior intended to hurt or destroy.
 - In United States, they are MUCH more likely to be murdered compared to most other developed nations.
- **The Biology of Aggression.**
 - Genetics
 - Neural Influences (is aggression in the brain)
 - Biochemical
- **The Psychology of Aggression.**
 - Frustration-Aggressive Principle: the blocking of an attempt to achieve some goal
 - Creates anger which generates aggression.
 - The goals can be sports or work, relationships, and body conditions.
 - Hot Weather and Aggression.
 - Aggression and TV.
- **Conflict** is a perceived incompatibility of actions, goals or ideas.
 - Social trap or prisoner's dilemma.
- **Attraction. 5 Factors of Attraction:**
 1. Proximity- geographic nearness.

2. Reciprocal Liking- You are more likely to like someone who likes you.
 3. Similarity
 4. Liking through Association- Classical Conditioning can play a part in attraction.
 5. Physical Attractiveness
- **Altruism** is unselfish regard for the welfare of others.
 - **Social Exchange Theory.** The idea that our social behavior is an exchange process, which we maximize benefits and minimize costs.

SOCIAL PSYCHOLOGY IN EVERYDAY LIFE

1. **Striving for mastery.** The motivational principle that people seek to understand and predict events in the social world in order to obtain rewards.
2. **Seeking connectedness.** The motivational principle that people seek support, liking, and acceptance from the people and groups they care about and value.
3. **Valuing me and mine.** The motivational principle that people desire to see themselves, and other people and groups connected to themselves, in a positive light.
4. **Conservatism.** The processing principle that individuals' and groups' views of the world are slow to change and prone to perpetuate themselves
5. **Accessibility.** The processing principle that the information that is most readily available generally has the most impact on thoughts, feelings, and behavior.
6. **Superficiality vs. Depth.** The processing principle that people ordinarily put little effort into dealing with information, but at times are motivated to consider information in more depth.

PERSONALITY

DEFINITIONS

1. Consistency - All individuals tend to behave, think, and feel in certain ways that is fairly consistent across time and situation. While some things change, other things remain very stable.
2. Personality - Personality is generally defined as an individual's unique and relatively stable patterns of behavior, thoughts, and feelings.
3. Behavioural Position - A strict behaviourist would argue against the existence of personality. They argue that behavior is determined by the situation you are in and consistency in behavior can be explained by consistency of situation.
4. Personality and Situations - Most psychologists would argue that there is an interaction between personality traits and situations. Cheating for example. Would you cheat if you were in the right situation and you knew you wouldn't get caught?

FREUD'S THEORY OF PERSONALITY

1. Levels of Consciousness - Freud viewed what we were aware of at any time (conscious level) as only the tip of the iceberg. Beneath this level was the vast majority of our consciousness. The preconscious is just below the conscious level. This contains memories and stored knowledge which we are not immediately aware of but which we

can retrieve at any time. Beneath the preconscious and forming the largest portion of the human mind is the unconscious. These are thoughts, desires, and impulses of which we are not aware. Even though we are not aware of these, Freud saw these as controlling our behavior and it is the goal of psychoanalysis to get at these unconscious factors.

2. Three Components of Personality

(a) Id: Present at birth. Operates on the pleasure principle. It is there to serve your instincts and it wants immediate gratification.

(b) Ego: Develops later when start learning and thinking logically. Realize that sometimes you may have to delay gratification or find other ways to get it. Works on reality principle, tries to find realistic ways of gratifying instinct.

(c) Superego: This is the ideal moral part. Strives for perfection rather than pleasure or reality. Works through making you feel guilty or ashamed when you do badly. Develops about 5-6 years of age.

3. Stages of Psychosexual Development - Five stages of development. Centers around sex instinct. As you mature your sexual interest shifts to different erogenous zones. Each stage is centered on a particular erogenous zone.

(a) Oral Stage - (birth to 1 year) erogenous zone is the mouth. Erotic pleasure comes from sucking, biting, spitting, etc.

(b) Anal Stage - (1-3 years) This is when child gets toilet trained, sphincter muscles begin to mature. Anus is erogenous zone and defecation becomes method of getting erotic pleasure.

(c) Phallic Stage - (3-6 years) Genital area becomes erogenous zone and child gets sexual pleasure from masturbation. Develop strong sexual desire for parent of the opposite sex. Oedipus Complex: Boy develops a desire to sleep with their mothers and they develop jealous of father so they want to destroy them. When his castration anxiety becomes intense enough he resolves the conflict by identifying with father, act like him, and that way he can vicariously experience mother. Girls experience Electra Complex which is similar to boys except that the girl develops penis envy and desires the father in the hope he shares his penis with her. Resolution is not as clear as for boys, it just fades away when she realizes she can't have father but this leads to less developed superego because its development is not based on as much fear (castration).

(d) Latency Period - (6-12 years) Sex instincts are relatively quiet. This continues up to puberty.

(e) Genital Stage - (12+) now interested in sex with opposite sex. Underlying aim is reproduction.

OTHER PSYCHOANALYTIC VIEWS

1. Jung - Jung was a disciple of Freud but the two of them had a bitter falling out. Jung agreed with Freud on the importance of the unconscious but proposed another important part, the collective unconscious. This is the part of the unconscious that is shared by all human beings and it is passed along to new generations biologically. It contains archetypes which predispose us to view the world in certain ways. Two important archetypes are the anima and animus. The animus is the masculine side of females and the anima is the feminine side of males. When we look for a mate we look for someone who can project this hidden part of our personality. Jung also coined the

terms introvert and extrovert to describe individuals who are shy and reclusive, or talkative and sociable.

2. Karen Horney - Freud's was a very sexist theory. Horney could be considered one of the first feminists as she criticized Freud for his ideas on female sexuality. Her focus was more on social factors in shaping the developing child rather than sexual factors.

HUMANISTIC THEORIES

1. Overview - While specific theories have their differences, humanistic theories are all based on a much more optimistic view of human nature. They propose that all humans strive for personal growth and development.

2. Rogers' Self Theory - Carl Rogers proposed that all humans are striving to become fully functioning persons. These are psychologically healthy people who enjoy life to the fullest. Why aren't we all this way? Because a gap develops between self-concept (the beliefs we have about ourselves) and reality. This creates anxiety and to reduce this anxiety we do things like distort our perceptions of reality or deny reality. This leads to maladjustment and personal unhappiness. This happens because we grow up in an atmosphere of conditional positive regard, our parents approve of us only when we behave in certain ways. To overcome this, you need an atmosphere of unconditional positive regard where you know you will be accepted no matter what you do or say.

3. Maslow - Maslow's need hierarchy was already discussed but this is also considered a humanistic theory. The key here is self-actualization, the top of the hierarchy. Maslow saw psychologically healthy people as ones who have attained self-actualization. They have reached their true potential, they recognize their shortcomings as well as their strengths, and they are interested in maintaining their own standards which may not necessarily be societies.

4. Self-Disclosure - This is simply the act of revealing intimate information about oneself to another person. This forms the basis of many therapeutic approaches, especially humanistic approaches. Does it work? It does seem to help both psychological and physical health although it may be best if it is reciprocal, that is, the person you are disclosing to disclose back.

TRAIT THEORIES

1. Overview - These are theories that focus on identifying key dimensions of personality on which people may differ. Ask class for examples of traits.

2. Allport's Trait Theory - Allport divided personality traits into several major categories.

(a) Secondary Traits - These are least important and they exert a small and relatively specific influence on personality.

(b) Central Traits - These are more important and are the five to ten traits that best describe the individual. This is the uniqueness of an individual's personality.

(c) Cardinal Trait - This is a single, all-important trait that characterizes a few people. Not everyone has this but someone like Michael Milken may have one (greed).

3. Key Dimensions of Personality - Research has identified five robust dimensions that characterize personality.

(a) Extraversion - Social, talkative fun-loving to shy, silent, and cautious (introversion).

(b) Agreeableness - Good natured, gentle, cooperative, helpful to irritable, suspicious, and uncooperative.

- (c) Conscientiousness - Careful, self-disciplined, and responsible to careless, weak-willed, and unscrupulous.
- (d) Emotional Stability - Poised, calm, composed, and not hypochondriacal to nervous, anxious, excitable, and hypochondriacal.
- (e) Openness to Experience - Imaginative, sensitive, intellectual to insensitive, narrow, and crude.

MEASURING PERSONALITY

1. Objective Tests - These are generally self-report measures in which people are asked whether statements are true or false about themselves or whether they agree or disagree about something. Just like on intelligence tests, performance is compared to some norm group. The MMPI is the most common. It contains over 500 true-false questions.
2. Projective Tests - These are tests where you are shown relatively ambiguous stimuli and you respond to it by telling what you see or telling a story about it. The most common are the Rorschach which is an inkblot test and the Thematic Apperception Test (TAT) in which a picture is shown and you tell a story about it. The utility of these tests is not agreed upon. Many people believe they are worthless.

PSYCHOPATHOLOGY

DEFINING PSYCHOLOGICAL DISORDER

1. What We Call It - Psychological disorders, as a class, are called many things. We use the term psychopathology, mental illness, behavior disorder, emotional disturbance, etc. They all mean roughly the same thing although the severity of the disorder may vary greatly.

They share the following features however.

2. Distress - The person is experiencing some sort of negative reaction such as anxiety, depression, confusion, etc.
3. Atypical - Behavior deviates from what society believes is normal. They may react differently to normal stressors or engage in behaviors society considers deviant. This type of definition has led to some behaviors we may now not consider as mental illness to be labelled as pathology such as homosexuality. This was considered to be a mental illness until the 1970's.
4. Maladaptive - The disorder interferes with normal, everyday functioning. They may not be able to hold a job, have normal social interactions, or care for themselves.
5. Unacceptable to Society - The behavior is in some way viewed negatively by society, as either objectionable or unacceptable. Again, homosexuality fell into this category.

HISTORY OF PERSPECTIVES ON MENTAL ILLNESS

1. Possessed by Demons - The earliest view of mental illness was that people suffering from it were possessed by demons. People were actually treated by drilling holes into their skulls to let the demons out. This procedure was called trephining.
2. Medical Model - The next view of mental disorder held that mental illness, just as physical illness, was the result of biological or medical factors. This view is still widely held today and it is probably valid for many mental disorders. It did lead to more humane treatment for the mentally ill.

3. Psychodynamic Perspective - This is the Freudian perspective. It holds that mental disorders are the result of unconscious urges or impulses.
4. Behavioural Perspective - This perspective holds that abnormal behavior is learned just like normal behavior. You are the product of your environment and through modelling, and social and cultural factors you learn to behave in a maladaptive manner.

DSM-IV

1. DSM-IV - The American Psychiatric Association publishes the "bible" of psychiatric diagnosis. It is the Diagnostic and Statistical Manual of Mental Disorders. It is usually just referred to as the DSM-IV, meaning that the version we are currently using is the third edition revised.
2. Diagnosis - What the DSM-IV does is give you a listing of behaviors that are associated with each classification. In order to use a diagnosis, the individual must meet the specific criteria set forth in the manual. It makes no claims to etiology (usually, PTSD is an exception) nor does it propose treatment.
3. Constantly Revised - As knowledge and attitudes change, so does the DSM-IV. As I mentioned, homosexuality was once considered a mental disorder, it no longer is. Other conditions like PMS (premenstrual Syndrome) and PTSD have been added in more recent versions. Others change. For example, what we now refer to as ADHD (Attention Deficit hyperactivity Disorder) was called Minimal Brain Dysfunction in DSM-II and Attention Deficit Disorder with or without Hyperactivity in DSM-III. DSM-IV is returning to the DSM-III categorization.

ANXIETY DISORDERS

1. Anxiety - We have all experienced anxiety, it is part of everyday life. For anxiety to become a disorder it must be intense enough and of long enough duration to cause disruption in functioning. **There are different types of anxiety disorders.**
2. Generalized Anxiety Disorder - This is when the person is experiencing persistent and severe anxiety in most situations. In other words, they worry about everything and can't function. There is a constant fear, tension, and dread.
3. Panic Attacks - This is when an individual experiences sudden intense anxiety which temporarily renders them incapable of functioning. They are brief but intense but do not appear to be triggered by a particular object or event.
4. Phobias - A phobia is defined as an irrational fear of a particular object or event such as a fear of snakes, heights, etc. We talked about these before when we discussed classical conditioning because they are thought to be learned through classical conditioning. Everyone has fears but it does not become a phobia unless it interferes with normal functioning.
5. Obsessive-Compulsive Disorder (OCD) - An obsession is a thought that you can't get out of your mind. A compulsion is a ritualistic behavior that you engage in over and over again such as washing your hands over and over to get rid of germs. These are anxiety disorders because the obsession creates anxiety and the compulsion is your means to overcome anxiety. Both are maladaptive and disrupt normal functioning.
6. Treatment - Generalized anxiety disorders are usually treated with medication but phobias and OCD are usually treated behaviourally.

DISSOCIATIVE DISORDERS

1. Overview - These include disorders of thought in which you lose identity or memory.
2. Psychogenic Amnesia - Amnesia is a loss of memory while it usually happens due to some physical cause it is sometimes due to psychological causes such as experiencing trauma or extreme stress. When there is no physical cause we call it psychogenic amnesia. It usually involves a selective or localized forgetting but in rare instances it can involve a total loss of identity.
3. Multiple Personality - This is very rare, much rarer than you would imagine considering made for TV movies and defense arguments in criminal trials. This is when an individual possesses more than one distinct personality and they may not be aware of each other. This is still a controversial area and many people do not believe it actually exists.

SOMATOFORM DISORDERS

1. Overview - These refer to disorders in which the person experiences physical symptoms in the absence of a physical etiology. It could be paralysis, deafness, or other physical disorder.
2. Hypochondriasis - You have all heard the term hypochondriac. That is what this is. They have an extreme fear of illness and while there is no physical illness, they constantly think they have something or that something (like cancer) is about to appear. To be a true hypochondriac you must be thinking of this constantly and it must disrupt your life. The key here is that there is really nothing wrong with you, you just think there is.
3. Conversion Disorder - In conversion disorder there is an actual impairment but there is no physical reason for it. Individuals have experienced blindness, deafness, and paralysis for no apparent physical reason. There have even been people who have appeared pregnant when they weren't. This is usually the result of extreme stress or trauma.

SEXUAL DISORDERS

1. Sexual Dysfunction - These include disturbances in sexual desire, arousal, and ability to achieve orgasm. Disorders of desire include an aversion or a lack of interest in sex. Disorders of arousal include failure to achieve erection in males or lubrication or vaginal constriction in women. Disorders of orgasm include failure to, achieve orgasm or premature ejaculation in men. This does not include individuals who have these problems due to medical reasons.
2. Paraphilias - A paraphilia involves arousal only to inappropriate sexual objects. These include pedophilia which is desire to have sex with young children. Fetishes which is arousal to inanimate objects such as shoes, underwear, and even dead bodies. Transvestitism is an example of a fetish. Sadism and Masochism is when you get sexual pleasure from receiving or inflicting pain and humiliation. Exhibitionism and voyeurism are also pedaphilias.
3. Gender Identity - This is transsexualism in which you believe you were born with one sexual genitalia but your identity is with the other gender. It may result in a sex-change operation.

MOOD DISORDERS

1. Overview - These are also called affective disorders and they involve inappropriate affect such as depression or extreme elation.
2. Bipolar Disorder - This is also called manic-depressive and these individuals experience extreme mood swings. they vary from extreme depression to mania in which they are extremely excitable and energetic. They are usually treated with Lithium.
3. Major Depression - This is an overwhelming and persistent feeling of unhappiness and despair. They see life as hopeless and tend to withdraw and become lethargic. If a loved one dies you become depressed, that is normal. This is prolonged feeling and not the temporary reaction to an unhappy experience. It may even lead to suicide. Treatment is usually with medication combined with therapy.

PERSONALITY DISORDERS

1. Overview - These are when a person has some personality trait which is inflexible and causes disruption of their everyday functioning. They include deviant behaviors and thoughts.
2. Antisocial Personality Disorder - These people are dangerous to others and they include delinquency, lying, and criminal activity. They are generally aggressive, irritable, impulsive, and fail to take responsibility for their actions. They are sometimes referred to as sociopaths. More common in males.
3. Histrionic Personality Disorder - This involves exaggerated displays of emotions. Even trivial events are blown up out of proportion to what happened. Everyday activities become major life events and they don't see why others don't share their concerns. More common in females.
4. Avoidant Personality Disorder - This is characterized by extreme shyness, poor self-esteem, and an inability to take criticism. They must have total acceptance so to avoid rejection they stay away from normal social interactions.
5. Dependent Personality Disorder - This is also the result of low self-esteem and they show extreme dependence on others. They let others tell them everything they are to do. They make no decisions on their own.

SCHIZOPHRENIA

1. Overview - These most serious psychological disorders and involve a lack of contact with reality. It is sometimes called a thought disorder and it is characterized by the following.
2. Disorder of Thought and Language - They do not think or speak like others. You could say it is weird. They often create their own words and they are hard to understand because their communication may consist of a bunch of seemingly unrelated statements. They also frequently have delusions which are firmly held beliefs that have no basis in reality. One of the more common types is delusions of persecution where you think everyone is out to get you. There are also delusions of grandeur where you think you are someone famous or powerful. Finally, there may be delusions of control in which they think that some evil force or God is controlling their brains.

3. Disturbance of Perception - They may not perceive the world in the same way as others. Most common is hallucinations where they have sensory experiences that have no basis in reality. The most common are auditory hallucinations where they hear voices but they may also have visual hallucination where they see things that are not there.

4. Disturbance of Emotion - The third key symptom of schizophrenia is inappropriate or unusual mood reactions. They may laugh at something sad or cry at a joke or do either for no reason at all.

TYPES OF SCHIZOPHRENIA

1. Disorganized Schizophrenia - The most striking feature is a childlike behavior. They act like young children do by giggling, making faces, etc.

2. Catatonic Schizophrenia - This involves a disturbance in motor behavior. They may be totally immobile and sit or stand in a position for indefinite periods of times. They may also have a waxy flexibility. That is you can mold them like a doll and they will maintain that position.

3. Paranoid Schizophrenia - This is characterized by delusions of persecution. They think everyone is out to hurt them. This is the most common type of schizophrenia.

*****the end*****

References

Books:

- Ariola, Mariano (2009). General Psychology with Drug Education. Quezon City. C and E Publishing
- Carlson, Neil (2007). Psychology: The Science of Behavior, 6th Ed. USA: Allyn and Palon Publishing
- Passa, Michael (2007). Psychology: The Science of Mind and Behavior, 3rd Ed. USA: McGraw-Hill Education
- Gerrig, Richard (2008) Psychology and Life, 18th Ed. USA: Pearson Education Australia
- Baird, Abigail (2010). Think Psychology. USA: Prentice Hall

Online

- <http://webpace.ship.edu/cgboer/genpsysyl.html>
- <http://psychology.about.com/od/sensationandperception/ss/perceptproc.htm>
- <http://psychclassics.yorku.ca/James/Principles/prin2.htm>
- <http://wilhelmwundt.com/works-cited.php>

Prepared by:

Marlon P. Tuiza, LPT, DPA
LET Reviewer, GE: Social Science